

Desirable Airfoil Features for Smaller-Capacity Straight-Bladed VAWT

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ABSTRACT

In the small scale wind turbine market, the simple straight-bladed Darrieus type vertical axis wind turbine (SB-VAWT) is very attractive for its simple blade design. A detailed aerodynamic performance analysis was conducted on a smaller capacity fixed-pitch SB-VAWT. Brief analyses of the main aerodynamic challenges of this type of wind turbine were first discussed and subsequently the authors conducted further literature survey and computational analysis to shortlist aerodynamic characteristics of a desirable airfoil for a self-starting and better performing SB-VAWT. The required geometric features of the desirable airfoil to achieve the short listed characteristics were also discussed. It has been found out that conventionally used NACA symmetric airfoils are not suitable for smaller capacity SB-VAWT. Rather, it is advantageous to utilize a high-lift and low-drag asymmetric thick airfoil suitable for low speed operation typically encountered by SB-VAWT.

Key words: Vertical Axis Wind Turbine, VAWT, Straight-bladed, Darrieus

ABBREVIATIONS AND ACRONYMS

A	projected frontal area of turbine
APG	adverse pressure gradient
C	blade chord
C_d	blade drag coefficient
C_{do}	zero-lift-drag coefficient
C_l	lift coefficient
$C_{l,max}$	maximum lift coefficient
C_P	turbine overall power coefficient
C_Q	turbine overall torque coefficient
C_t	tangential force coefficient
D	blade drag force
H	height of turbine
$HAWT$	Horizontal Axis Wind Turbine
L	blade lift force
N	number of blade
R	turbine radius
RN	local Reynolds number = WC/u
$SB-VAWT$	Straight-bladed vertical axis wind turbine
SPL	Sound Pressure Levels

V_a	induced velocity
V_∞	freestream wind velocity
$VAWT$	vertical axis wind turbine
W	relative flow velocity
α	blade angle of attack
θ	azimuth angle
λ	tip speed ratio = $R\omega/V_a$
μ	aspect ratio = H/C
σ	solidity = NC/R
ω	angular velocity of turbine in rad/s

I. INTRODUCTION

In the small scale wind turbine market, the simple straight-bladed Darrieus vertical axis wind turbine (SB-VAWT) is very attractive for its simple blade design. The main advantage of this category of wind turbine is its single moving part (the rotor) where no yaw mechanisms are required, thus simplifying the design configurations significantly. Blades of SB-VAWT may be of uniform section and untwisted, making them relatively easy to fabricate or extrude, unlike the blades of horizontal axis wind turbine (HAWT), which should be twisted and tapered for optimum performance. Furthermore, they produce less noise than the HAWT with the same power coefficient at normal operating speed [1] and almost all of the components of SB-VAWTs requiring maintenance are located at the ground level facilitating the maintenance work appreciably.

Environmentally benign SB-VAWTs can be utilized for a range of applications, including (i) electricity generation; (ii) pumping water; (iii) purifying and/or desalinating water by reverse osmosis; (iv) heating and cooling using vapour compression heat pumps; (v) mixing and aerating water bodies; and (vi) heating water by fluid turbulence. In general, SB-VAWT can sensibly be used in any area with sufficient wind, either as a stand-alone system to supply individual households with electricity and heat, or for the operation of freestanding technical installations.

Selection of airfoil is the most critical factor in achieving better aerodynamic performance and in determining the optimum dimensions of a SB-VAWT. Airfoil related design changes also have the potential for increasing the cost effectiveness of VAWTs [2]. All the earlier research works carried out by Sandia National Laboratory, Natural Resources Canada and several commercial SB-VAWT models used mainly symmetric airfoils, that were developed for the aviation applications, but were unable to self-start properly. If this problem of self-starting could be overcome without a big increase in cost, SB-VAWTs could make a significant contribution in the field of stand-alone power supply systems [3].

Against this background, this paper aims to perform detailed systematic investigative analysis with recently developed asymmetric airfoils appropriate for self-starting and better performance of smaller capacity fixed-pitch SB-VAWT through critical examination of their salient aerodynamic factors. In the next section, a brief overview of the main aerodynamic challenges of SB-VAWTs is presented. Realizing these challenges in mind, literature survey and computational analysis were conducted to identify and shortlist the desired aerodynamic characteristics of the airfoil suitable for smaller-capacity self-starting SB-VAWT with optimum performance. Finally, desirable geometric features of an airfoil suitable for smaller capacity SB-VAWT are discussed.

2. MAJOR AERODYNAMIC CHALLENGES OF SMALL-SCALE SB-VAWT

The smaller-capacity Fixed-pitch SB-VAWT is a lift force driven wind turbine consisting of two or more airfoil shaped blades which are attached to a rotating vertical shaft as shown in Figure 1. The wind blowing over the airfoil contours of the blade creates aerodynamic lift and pulls the blades along.

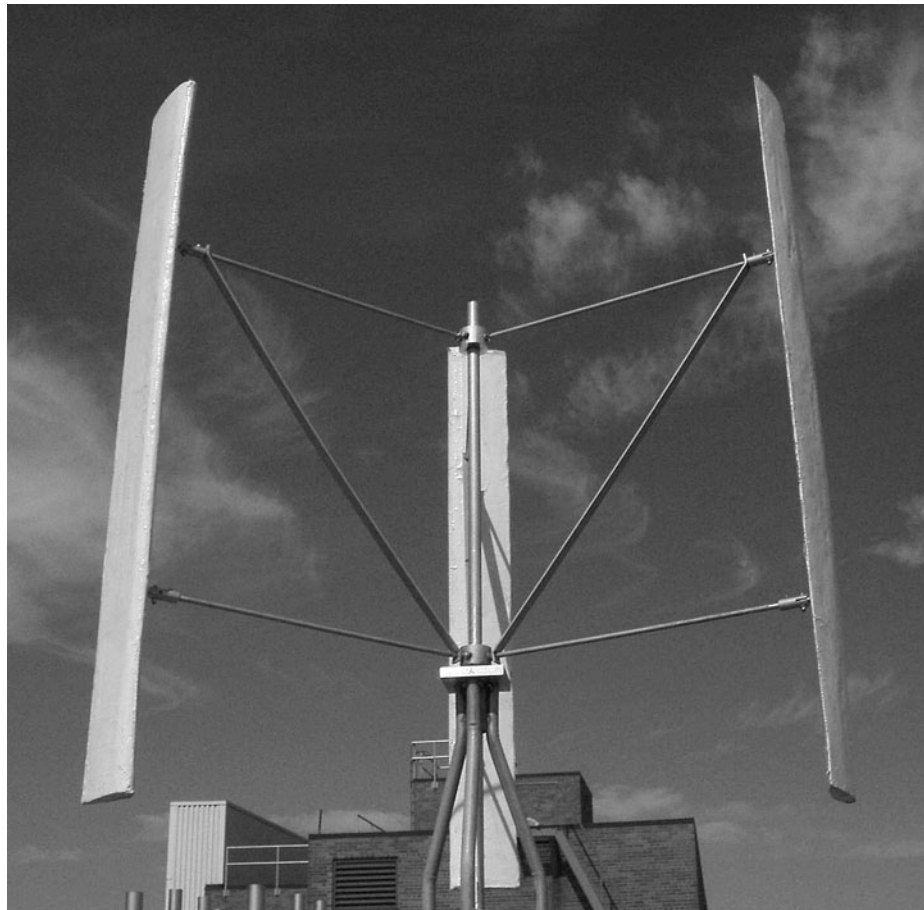


Figure 1. A Roof-top SB-VAWT Prototype

In Figure 2, a SB-VAWT blade's circular path is shown with both the 3D & 2D representations. As this blade rotates, it experiences a varying relative flow velocity (W), which is the vector sum of the local induced wind speed and the blade speed. The lift force generated by the blade has a tangential component in the direction of rotation. If the contribution of the drag force is smaller than that of the lift force, the blade contributes positive torque which drives a load connected with the central rotating shaft. The orbital position of the blade is called azimuth (θ) and both relative flow velocity and the local angle of attack (α) vary with θ . The amplitude of this variation is related to a non-dimensional parameter known as tip speed ratio ($\lambda = (R\omega)/V_a$). The variation of α with θ at different λ is shown in Figure 3 for a SB-VAWT with NACA 0015. It can be seen from this figure that α changes through positive and negative values in each revolution of the blade at different λ and as the turbine speed increases, more energy is extracted and the flow is decelerated.

Though SB-VAWT has very simple structural features, its rotor aerodynamics is very complex. The major aerodynamic challenges faced by smaller capacity fixed-pitch SB-VAWT are:

- They operate in the low Reynolds number (RN) regime which is a highly sensitive unstable region with high probability of separation. Considering the chord length as the characteristic length, the operating chordal RN ($\rho V_\infty c / \mu$) of interest is restricted between 100,000 to 500,000, which is considered as Low RN [4]. In this range of RN, very complex flow phenomena take place within a short distance of the leading edge on the upper surface of the blade [5] and the laminar separation bubble that commonly forms in this range plays an important role in determining the boundary layer behaviour and the stalling characteristics of the blade [6, 7].

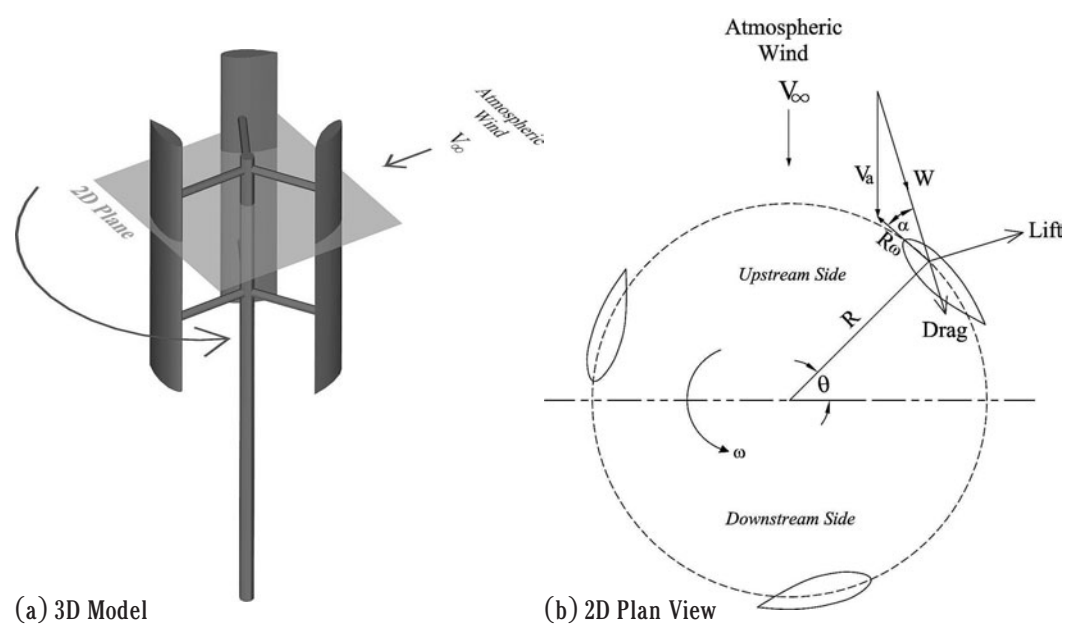


Figure 2. Flow Velocities around SB-VAWT

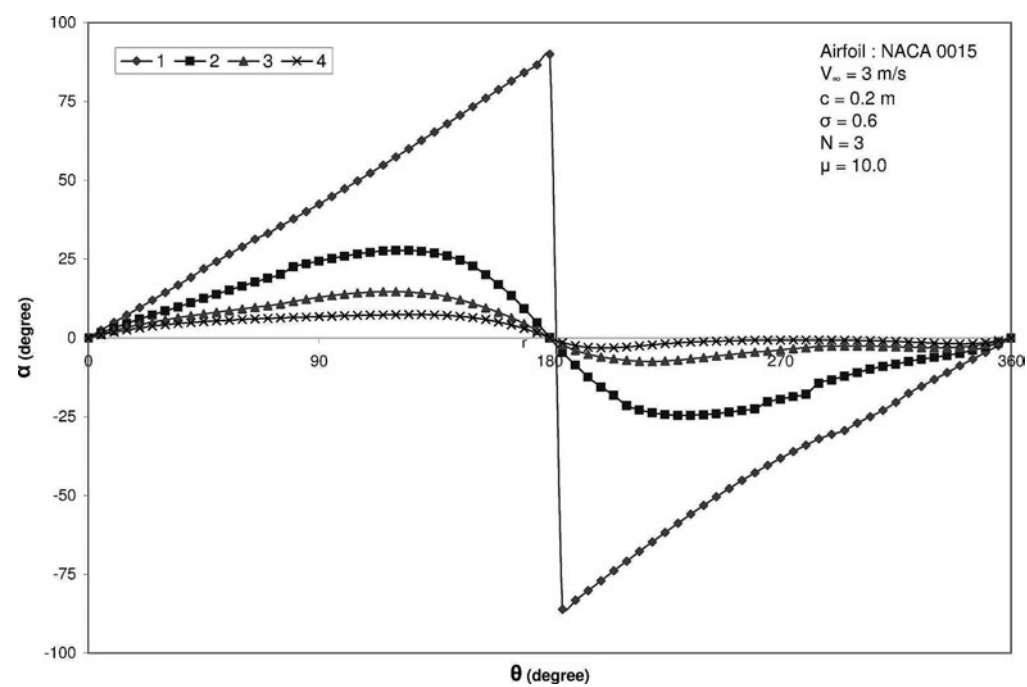


Figure 3. Variation of Angle of Attack and Azimuthal Position at different Tip-Speed Ratios of a SB-VAWT

- Because of the oscillating α , the blades always produce a fluctuating force, even in steady conditions. Consequently, undesirable dynamic stall, vibrations, gust response etc. may result, where the most important one is the dynamic stall effect which is dependent on parameters including airfoil shape, amplitude and oscillation frequency of angle of attack, type of motion, turbulence-level and three-dimensional effects.
- They operate in circular motion and thereby encounter flow-curvature effects if the chord/radius ratio is high which can have a detrimental influence on the performance of a SB-VAWT [8], especially at higher λ [9, 10].
- Unlike the conventional aerodynamic applications, VAWTs encounter a wide range of angle of attack, especially at low tip speed ratios as shown in Figure 3. When the turbine starts and as the rotational speed increases, the blades can even experience back flow [11]. In this situation, the effective angle of attack can be greater than the stall angle and thus the air flow detaches from the low pressure side of the blade airfoil [12]. As α over the blade is increased, at some point the air will separate, which usually starts at the trailing edge and shifts forward with increasing α . This phenomenon is called deep stall and has a strong detrimental influence on the performance of SB-VAWT blade mainly because of a hysteresis loop where the lift is small while drag remains large [11].
- The blades on the downstream-pass operate in their own wakes shed by blades on the upstream pass; for all the VAWTs, the blade/blade wake interaction presents the most fundamentals modelling problem [13]. It has been found out that most of the power is extracted from the wind during the upstream side of a VAWT [3].
- They suffer from parasitic drag losses due to radial arms which support the blades and transform the kinetic energy of wind into the rotating shaft. The parasitic drag losses reduce the overall power output.
- They mostly operate in turbulent atmospheric conditions. As per a detailed computational analysis, their performance can be reduced significantly due to atmospheric turbulence [14].

All these factors collectively make the thorough analysis of straight-bladed Darrieus VAWT a challenging undertaking. One of the most problematic aspects of the complex aerodynamics of SB-VAWT is that they produce very little starting torque when conventional airfoils are used.

3. SELF-STARTING PROBLEM OF SB-VAWT

It is commonly believed that small-capacity SB-VAWT are inherently unable to self-start properly. This notion is true for older designs which were constructed by utilizing old symmetric NACA airfoils. According to Watson [15], the inability of Darrieus type VAWTs to self-start is due to a band of tip speed ratios below operating condition for which the net amount of energy collected by each blade in each revolution is negative. To achieve self-starting, SB-VAWT must be altered so that a net positive amount of torque is produced at all tip speed ratios up to the operating point. As shown in Figure 3, at low values of λ , the blade of SB-VAWT encounter a wide range of α and thus fluctuates between stalled and unstalled conditions. At low λ , the blade is stalled at almost all azimuth angles. As a result, most fixed pitch Darrieus turbines do not self-start reliably and it is only at higher values of λ , above about 3, that the blades remain 'unstalled' and the turbine can achieve high efficiency [3].

The inability of SB-VAWTs to self-start can be better understood from Figures 4 and 5, which illustrate the power (C_P) and torque (C_Q) coefficients of a 1 kW SB-VAWT at 3 m/s. The cascade model, proposed by Hirsch and Mandal [16] was utilized for producing these curves. The original source of the aerodynamic characteristics of the airfoil used here is Sheldahl and Klimas [17] which has been synthesised from a combination of experimental results and computer calculations by Lazauskas [18] for eliminating data anomalies. Both of these curves depict negative C_P and C_Q values at low tip-speed ratios. These negative regions were described by Baker [19] as the “dead band” which is an undesirable phenomenon. A turbine with this kind of C_P - λ characteristic may just start, but will not generally accelerate through the dead band to its optimum working speed unless the wind speed happens to drop suddenly in such a way that λ jumps from a value below the dead band to a value above it, and then increases gradually so the turbine can accelerate up to working speeds [3]. Kentfield [20], who developed a form of self-starting variable pitch VAWT, states in a recent publication that - “The Darrieus turbine...is inherently non self-starting” after realizing this undesirable feature of SB-VAWT.

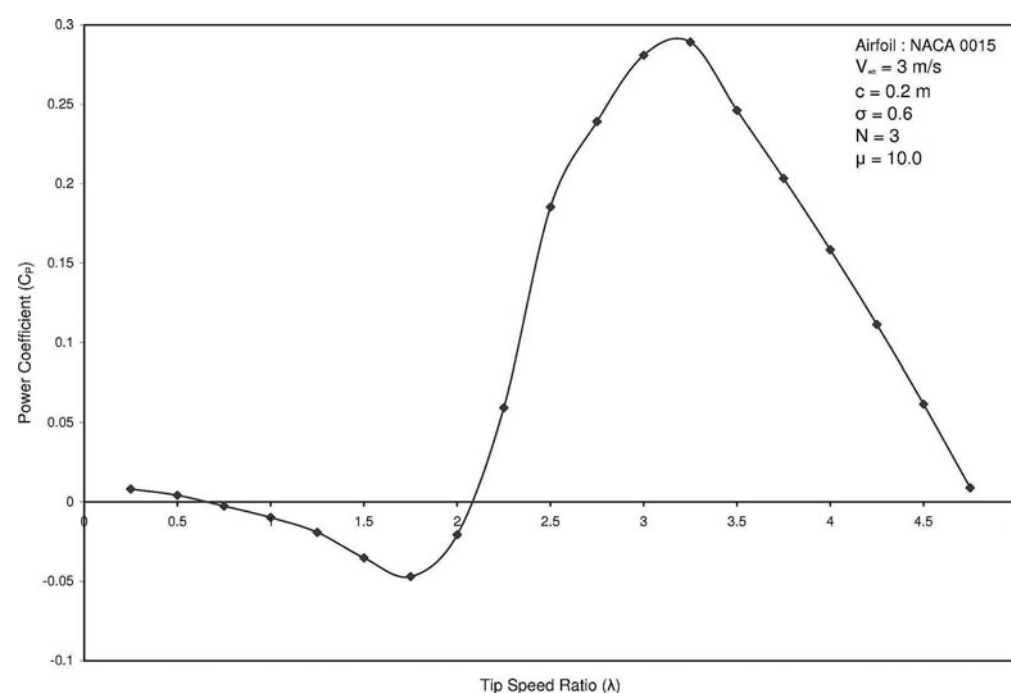


Figure 4. C_P - λ Curves of SB-VAWT with NACA 0015 and 0018 Airfoil Shapes

The problem of self-starting can be alleviated by (i) using high-lift low-drag special-purpose airfoil; and (ii) by incorporating a Savonius rotor or torque tube [21]. Self-starting problems can be overcome electrically if the SB-VAWT is coupled with a synchronous alternator or a DC generator which can function as a motor and can be used to drive the turbine up to operating speed. Although motor starting is used in grid-connected wind turbines and could be done with smaller capacity wind turbines, it is not a common practice. A detail investigation of self-starting features of SB-VAWT was performed by Kirke [3] with cambered blades using profiles that exhibit superior characteristics at low Reynolds numbers and his results indicate that fixed pitch VAWTs using these blade sections should self-start reliably.

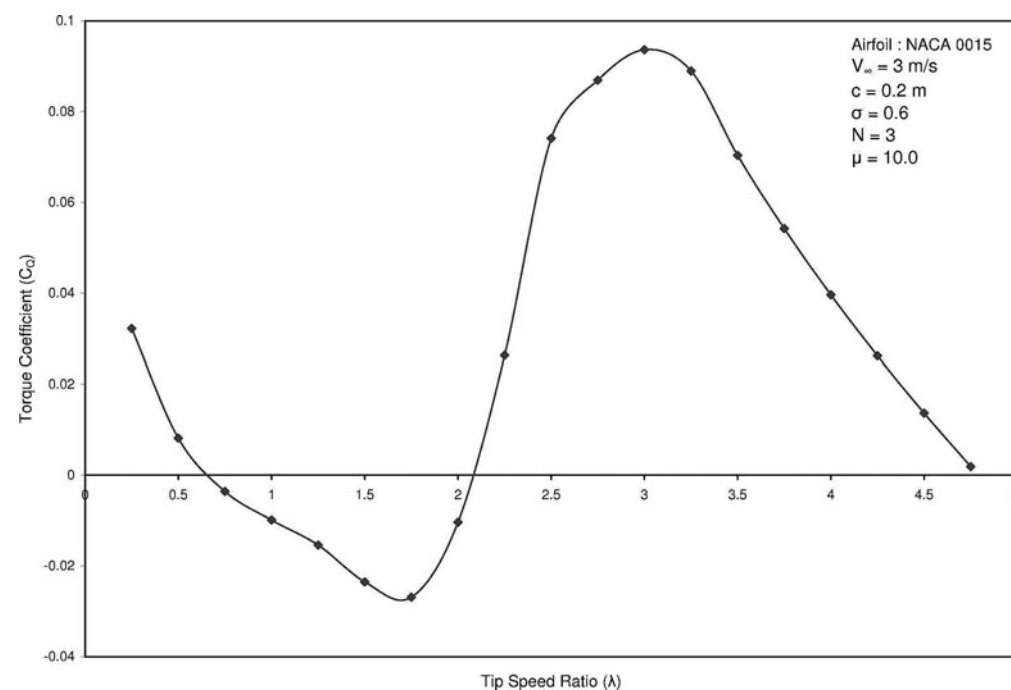


Figure 5. C_Q - λ Curves of SB-VAWT with NACA 0015 and 0018 Airfoil Shapes

4. PREVIOUSLY UTILIZED GENERAL-PURPOSE AIRFOILS FOR SB-VAWTS

Since its early stage of design and development, both the straight-bladed and curved-bladed Darrieus type VAWTs were mostly made with blades of the symmetric airfoils from the NACA 4-digit series, mostly NACA 0012, 0015, 0018 as shown in Figure 6, which were originally developed for high Reynolds number aviation applications. These airfoils have presumably been used mainly because only for these airfoils was there sufficient information on performance and loading [22], including low RN data. Their lift, drag and pitching moment characteristics are the most well documented and thus making validation of the theoretical predictions easier [3]. However, Migliore [23] has shown that the use of NACA 6-digit airfoils can improve the performance of the curved-blade VAWTs operating at Reynolds number of 3×10^6 . But, their application at lower RN is yet to be investigated. In 1980, Sandia National Laboratory (the main organization devoted in VAWT related research and development works in the U.S.A) developed a family of airfoil sections specifically tailored for use in the Egg-beater type Darrieus VAWTs [2], but like NACA 6-digit airfoils, suggested by Migliore, their application for smaller capacity VAWTs is yet to be analyzed.

In the late seventies, Healy [24] performed a theoretical study of the performance of VAWTs with NACA 0018 and eight cambered airfoils (namely Gö 460, Gö 676, Gö 738, Gö 735, Gö 746, Gö 741, Gö 744 and Gö 420). From his investigation, Healy concluded that “the closer the airfoil is to symmetric, the more satisfactory its power output.” Baker [19] found satisfactory performance with the moderately cambered Gö 420 (also known as GOE 420) which is described below. Furthermore, Kirke [3] questioned Healy’s conclusion by putting forward several valid arguments in favour of cambered airfoils for self-starting SB-VAWT. In the mid-nineties, Kirke [3] performed detailed computational works related to fixed-pitch SB-VAWT and he argued that – “for a small fixed pitch VAWT to self-start it is necessary that the aerofoils produce at least moderately high lift and low drag at Reynolds number between about 80,000 and 150,000. The symmetrical sections usually used on VAWTs perform poorly in this RE range

and do not achieve self-starting, but thin cambered sections perform much better – at least at positive incidence. Unlike symmetrical sections, thin cambered sections can produce moderate lift at positive incidence even with a totally laminar boundary layer, i.e. at very low Re [25]. Although cambered sections at negative incidence develop little lift and have a low stall angle, it is possible that their much superior performance at positive incidence could result in superior overall performance at low Re . Subsequently, Kirke performed simulation with NACA 4415 and one of the recently developed asymmetric airfoils (S1210) and commented that – “performance of fixed pitch VAWTs with cambered blades using newly developed profiles that exhibit superior characteristics at low Reynolds numbers. Results indicate that fixed pitch VAWTs using these blade sections should self-start reliably”. However, Kirke also found out that the S1210 blades would cost no more to produce than NACA 0015 blades and their only drawback for high speed loads would be the fact that maximum C_p occurs at lower λ with S1210 blades than with NACA 0015 blades.

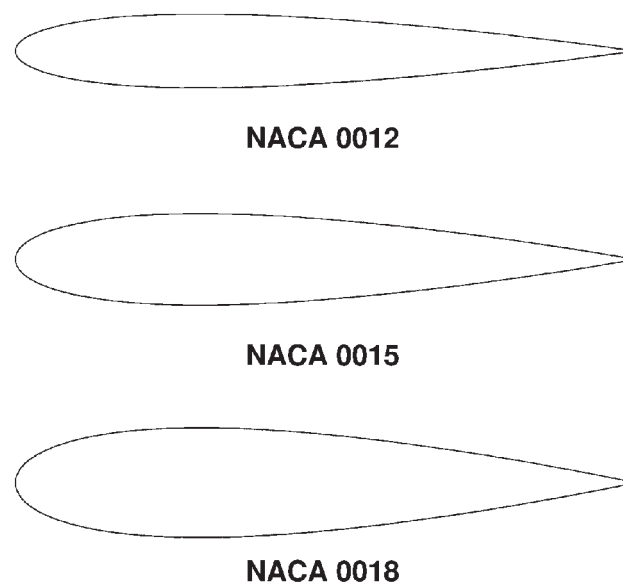


Figure 6. Geometry of Conventionally used NACA Symmetric Airfoils

In the early eighties, Kadlec [26] of Sandia National Laboratory suggested that the power coefficient can be increased by using cambered airfoils on blades with little or no cost increases and he envisaged that the continued investigation of these alternative airfoils will determine their feasibility in advanced VAWTs. It was also acknowledged by Klimas [2] that cambered blades appeared to offer some advantages in VAWT design.

Zervos [27, 28] evaluated six airfoils (namely NACA 0012, NACA 0015, NACA 0018, NACA 63-015, GAW-1 and NLF0416) and found that the results of instantaneous power coefficient, normal force coefficient, and tangential force coefficient show a net advantage for GAW(1), also known as LS-0417, and NLF0416 both in pressure distribution (smaller adverse pressure gradients) and instantaneous loading (a more symmetric distribution with smaller peaks of normal and tangential forces during the cycle). Based on these results, he suggested that “judicious use of non-symmetric airfoils could produce better performance characteristics”.

Baker [19] investigated the cause of the inability of low solidity fixed-pitch SB-VAWT to self start and suggested the utilization of blades having airfoil sections that strongly exhibit the

laminar separation bubble phenomena below the stall and that have low immediate post-stall drag. He investigated the performance of NACA 0012, Gö 420 and Wortmann FX63-137. He found superior performance of Gö 420 in comparison to NACA 0012 and the performance of FX63-137 was unsatisfactory. Baker commented that the inferior performance of FX63-137 is due to its very sharply angled trailing edge which exhibited noisy stall from positive angles of attack, accompanied by a sudden deceleration. The inferior performance of FX63-137 can also be attributed to its large camber which can be counterproductive. It was commented by Dereng [29] that excessive camber or lack thereof will eliminate initial start-up capability and severely degrade or eliminate the ability to traverse the intermediate tip speed ratios.

It should be added that, cambered airfoils can be attached with the supporting struts of fixed-pitch SB-VAWT in two configurations based on the concave curvature of the camber line. These configurations are: (i) Concave-out; and (ii) Concave-in configurations as shown in Figures 7(a) and 7(b) respectively. If the concave side of a cambered blade faces outwards (i.e. the Concave-out configuration), it experiences positive incidence on the upstream pass and negative incidence on the downstream side as shown in Figure 7(a). As SB-VAWTs extract more energy on the upstream side, reduced amount of energy is available on the downstream side and as a result α for an unpitched blade is less on the downstream side than on the upstream side [3]. According to Kirke [3], this configuration ensures maximum power extraction from the undisturbed flow on the upstream side. When the concave side of the cambered airfoil faces inward (i.e. Concave-in configuration), negative incidence occurs on the upstream side and positive incidence occurs on the downstream side as can be seen from Figure 7(b). For SB-VAWTs with large chord-to-radius ratio, the concave-out configuration results in addition of the virtual camber with the actual camber due to flow curvature effects. On the other hand, virtual camber for concave-in configuration will tend to cancel out the actual camber.

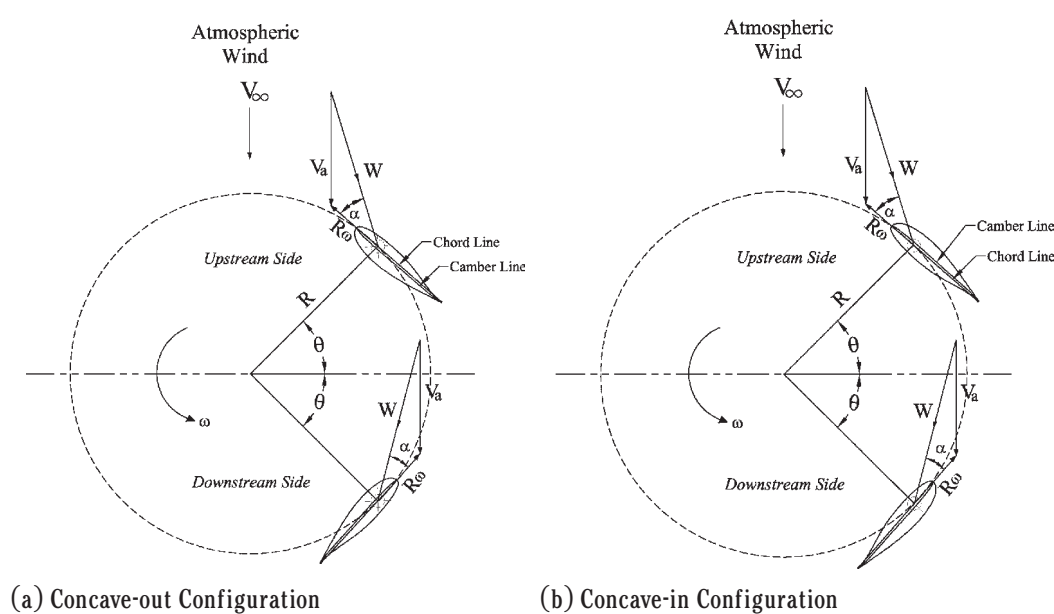


Figure 7. Two types of configurations for attaching cambered airfoils with the supporting struts

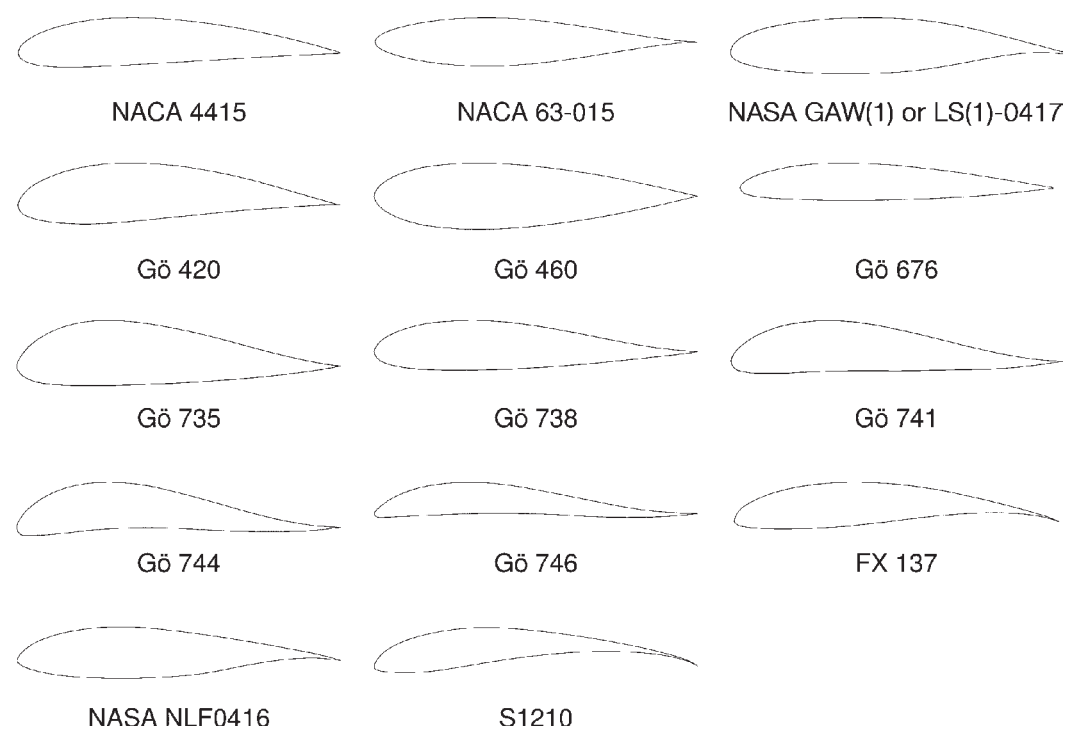


Figure 8. Geometries of Selected Asymmetric Airfoils

4.1. Comparison of Performance of Selected Asymmetric Airfoils

Based on the literature survey, only the prospective airfoils (as shown in Figure 8) which are available in the public domain are investigated in this section to compare their performance. The selected airfoils are (i) Göt 420 or GOE 420 [17], (ii) NACA 4415 [3] (iii) LS(1)-0417 or commonly known as GAW-1 [27], (iv) NLF(1)-0416 [27], and (v) S1210 [3]. The Reynolds number of the present analysis is limited between 100,000 and 300,000 which is the typical operating condition of smaller-capacity SB-VAWTs. Performance of these five selected airfoils is also compared with the NACA 0015 which has been extensively applied as a blade shape in the past.

In Figures 9 and 10, the tangential force coefficient (C_t) versus angle of attack curves for 5 selected asymmetric airfoils are compared with NACA 0015 at $Re=100,000$ and $300,000$ respectively. For obtaining these curves, XFOIL [30] was used to generate the required 2D datasets of lift and drag coefficients at $-20^\circ \leq \alpha \leq 20^\circ$. XFOIL is an interactive program developed by MIT Professor Mark Drela [31] for the design and analysis of subsonic isolated airfoils and it has been widely utilized for low Re airfoil analysis by renowned wind turbine related research organizations including NREL [32], Risoe [33, 34] and Delft University [35].

Section lift and drag data at high angles of attack (i.e. the post-stall regime) are not available for many airfoils of interest to wind turbine analysts [37] and in this regime the flow becomes less dependent on the airfoil shape and behaves similar to high angle of attack flow around flat plates. Using this notion, the models (like Viterna-Corrigan model) have been developed that are commonly used to extrapolate the available airfoil data across the entire 360° range of potential angles of attack [38]. The Viterna-Corrigan model can be used to predict the lift and drag coefficients in the post-stall regimes of any airfoil just by using only a few pieces of experimental or computed data in the pre-stall regime [39]. For the present analysis, the FoilCheck program of NWTC [40] has been utilized to generate different airfoil

characteristics between $\alpha = \pm 180^\circ$ using the computational 2D results obtained from XFOIL around $-20^\circ < \alpha < 20^\circ$. FoilCheck uses a combination of wind-tunnel data, the Viterna equations for deep stall, and user experience to generate airfoil data for all angles from a limited set of measurements [41]. However, it should be mentioned that this is a simplistic approach and in reality the situation might be more complex. The effects of free stream turbulence, as stated at the end of Section 2, are typically detrimental [42].

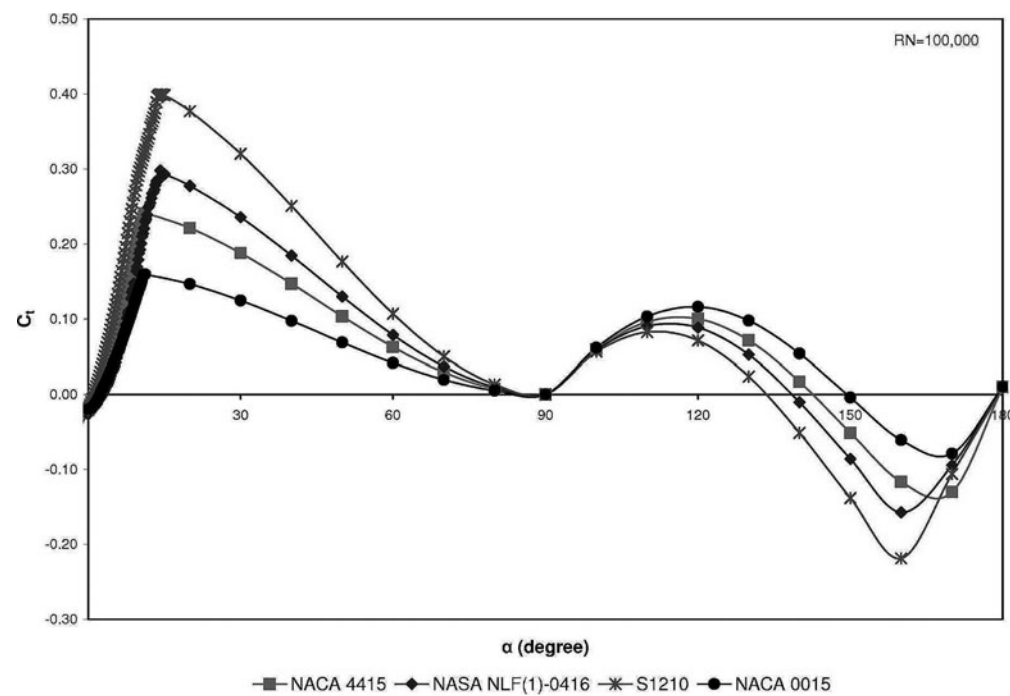


Figure 9. C_l - λ curves of Selected Asymmetric Airfoils at $RN=100,000$ & $\alpha=10$

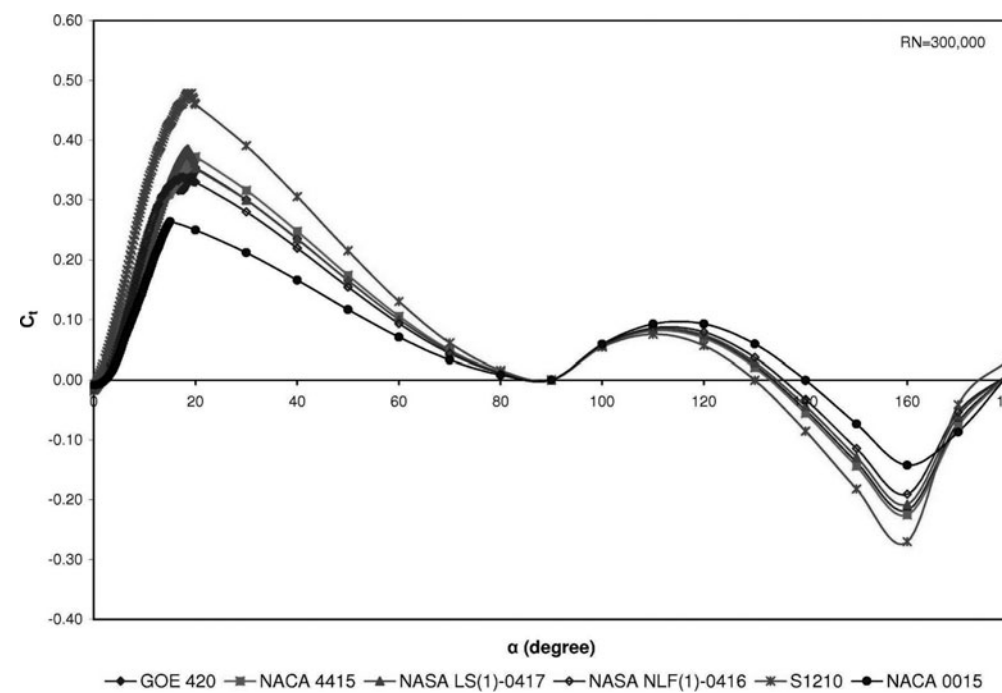


Figure 10. C_l - λ curves of Selected Asymmetric Airfoils at $RN=300,000$ & $\alpha=10$

For SB-VAWTs, when C_t value of an airfoil shape is positive, the blade will produce forward tangential or thrust force. Instantaneous torque produced by the blades can be calculated when the thrust force is multiplied by the blade radius (R). It can be seen from Figure 9 that C_t values of NACA 4415, NLF(1)-0416 and S1210 are much superior than that of conventional NACA 0015 at $RN=100,000$ for $0 \leq \alpha \leq 90^\circ$ degrees. However, for $90^\circ \leq \alpha \leq 180^\circ$, the C_t values of NACA 0015 are greater than the asymmetric airfoils. Due to convergence problem, the aerodynamic characteristics of Gö 420 could not be generated from XFOIL at $RN=100,000$, possibly due to laminar separation bubbles typically encountered by airfoils at low RN . As a result, C_t values of these two airfoils could not be shown in Figure 9. For $RN=300,000$, there was no problem of convergence and lift and drag coefficients could be generated for all the asymmetric airfoils and their corresponding C_t values are shown in Figure 10. This figure also shows the superiority of asymmetric airfoils over NACA 0015 for $0 \leq \alpha \leq 90$ degrees. Among all the asymmetric airfoils, the performance of S1210 is the best for $0 \leq \alpha \leq 90$ degrees and thus confirming Kirke's [3] claim that a fixed pitch SB-VAWT using this low RN airfoil should self-start satisfactorily.

5. SPECIALLY DESIGNED AIRFOILS FOR SB-VAWTS

Rather than applying existing airfoils designed for aviation industry, several researchers have attempted to explore different types of asymmetric airfoils (mainly by modifying the existing airfoils) and claimed that they perform better than the conventional airfoils designed for the aviation applications. Kato et. al. [43], Dereng [29] and Seki [44] utilized special-purpose asymmetric airfoils in their US patented SB-VAWTs. From literature survey, four other special-purpose asymmetric airfoils could also be found which were developed for SB-VAWTs. These four airfoils are briefly described below.

5.1. "TWT Series" of Tokai University, Japan

In the early eighties, Seki et. al. [45] and Kato et. al. [46] developed a family of asymmetric airfoils for SB-VAWT that are called TWT series which had partly positive and partly negative camber. They utilized the single streamtube model to develop this series of asymmetric airfoils. The researchers claimed that these airfoils are better performing than the NACA 4-digit series and they presented a performance comparison of one of their airfoils with NACA 0012. Subsequently, they constructed two prototypes for electricity generation and heating of a greenhouse [46, 47]. However, these prototypes are equipped with a Savonius rotor and a starter motor, indicating their inability to self-start. No information regarding the coordinates or experimental datasets of these TWT series airfoils could be found from the literature.

5.2. "ARC Series" of University of Athens, Greece

In the late eighties, Zervos of University of Athens [48, 49] developed the Arc series specifically intended for VAWT. This series was basically developed by modifying the NACA 4-digit airfoils (like NACA 0015 or 0018) which were essentially designed for high speed applications. Like the TWT series, the coordinates or experimental datasets of these airfoils could not be found from the literature. Also, Zervos et al. did not present any discussion about the self-starting characteristics of the experimental fixed-pitch SB-VAWTs they made out of these airfoils and their plots do not include the behavior of these experimental models at low tip speed ratios [48-50].

5.3. “Transformed NACA 0018” of Maeda Corporation & Tokai University, Japan

Recently, Maruyama et. al. [51] have attempted to utilize a special-purpose airfoil by transforming existing NACA 0018 by curving the chord line along the rotating circumference. They fabricated a full-scale model with this airfoil and conducted experimental investigations. They found that the performance of the transformed NACA 0018 is better than the unmodified one under no-load condition. No information could be found regarding the starting characteristics of the full-scale model.

5.4. DU 06-W-200 of Delft University:

Claessens [11] conducted a design process under the scope of his Masters thesis. His design process was developed with the purpose of improving the NACA 0018 airfoil. He also investigated VAWT aerodynamics to find the design goals for the airfoil characteristics. At the end, he designed a special-purpose VAWT airfoil named “DU 06-W-200” which is a laminar, 20% thick airfoil with 0.8% camber. To be able to compare the performance of NACA 0018 and DU 06-W-200, wind tunnel measurements were performed at the Delft University of Technology. It was found by Claessens that (i) The 20% thickness adds to the blade strength without decreasing the performance; (ii) The added camber of 0.8% increases the performance with respect to a symmetric airfoil; (iii) The DU 06-W-200 performance equals the NACA 0018 for negative angles of attack; (iv) The DU 06-W-200 has a much higher $C_{l,max}$ for positive angles of attack, resulting in a wider drag bucket; (v) Deep stall occurs at higher angles of attack with a smaller drop in lift coefficient; (vi) In contrast to the NACA 0018 the DU 06-W-200 does not have laminar separation bubbles which extend over the trailing edge; (vii) The increase in turbine performance at the operating tip speed ratio of $\lambda = 3$ is 8% in clean conditions and twice as much when dirty.

Based on the published documents, the following points can be inferred regarding the above-mentioned four dedicated airfoils for SB-VAWTs:

- All of these airfoils are asymmetric airfoils;
- The main emphasis was to improve the overall performance of the turbine (at higher tip speed ratios) rather than developing a self-starting SB-VAWT.
- None of these airfoils were designed for the low RN (between 100,00 and 300,000) operation typically encountered by smaller capacity SB-VAWTs;
- Except the ARC series and DU 06-W-200, the other airfoils have not been designed considering the unsteady (dynamic stall) and rotational effects (flow-curvature).

6. DESIRABLE AERODYNAMIC CHARACTERISTICS OF THE AIRFOIL

In this section, an attempt will be made to short list the desirable aerodynamic characteristics of a self-starting and optimum performing SB-VAWT based on previous researches and computational analysis. The asymmetric airfoils analyzed in Section 4.1 are investigated using XFOIL. The airfoils described in Section 5 could not be incorporated in the analysis due to non availability of coordinates and copyright related issues. The analysis of the performance of the asymmetric airfoils is performed on different aerodynamic characteristics for Reynolds number 100,000 and 300,000. It should be pointed that it was not possible to generate the aerodynamic coefficients for GOE 420 at RN=100,000 due to convergence problem. However, at higher RN of 300,000 there was no such problem of convergence with any of the airfoils investigated in this section.

6.1. Large Stall Angle at Small Reynolds Number

It was discussed earlier that, behaviour of lift and drag coefficients in the post-stall conditions at low λ has serious consequence for fixed-pitch SB-VAWT and their lack of starting torque is due to the cyclical change in α with θ . As at low λ , α exceeds the stall angle for much of the blades' travel path. Thus stalled blades generally contribute negatively to the driving torque so that the net work output per revolution may be negative for some values of λ . If the stall angle can be increased, this situation will be improved as the blades are stalled for a smaller proportion of their travel. So, it is clear that the stall angle of the airfoil sections for fixed-pitch SB-VAWT should be as large as possible in the low RN operation, typically encountered by SB-VAWT, in order to minimize separation at low tip speed ratios [3]. In Figures 11 and 12, the lift curves of the selected prospective airfoils are shown at RN=100,000 and 300,000 respectively. It can be seen from Figure 11 that at positive incidence stall angles (α_{stall}) of S1210 is best, and it is followed by NACA 4415, NLF-0416 and LS(1)-0417 at RN=100,000. Figure 12 shows that α_{stall} of all the five prospective airfoils are similar at RN=300,000 and higher than NACA 0015 which justifies their superior performance in the positive incidences.

It is also evident from Figures 11 and 12 that performance of NACA 0015 is better than the other asymmetric airfoils at RN of 100,000 and 300,000 in the negative incidences. But, as illustrated in Figure 3, SB-VAWT airfoils encounter negative incidences in the downstream side ($180^\circ \leq \theta \leq 360^\circ$) where lesser amount of energy is available for harnessing as most of the power is extracted from the wind during the upstream side of a VAWT [3]. The vortex flow model of Duremberg [52] finds that the velocity difference across a three bladed turbine is about 0.55, i.e. the downstream blades receive wind at a speed of 0.55 times of the upstream side. About 90 or 95% of the energy is extracted from the upstream pass and this proportion depends on various factors including α and λ [3]. Loth and McCoy [53] have shown that a trade-off exists between energy extraction on the upstream and downstream passes and the more power is extracted on the upstream pass, the less energy is available on the downstream pass. So, though asymmetric airfoils produces lesser amount of lift in comparison to the symmetric airfoils at negative incidences and have a lower stall angle, their better performance at positive incidence can result in superior overall performance at low RN if the concave out configuration (shown in Figure 7) is used [3].

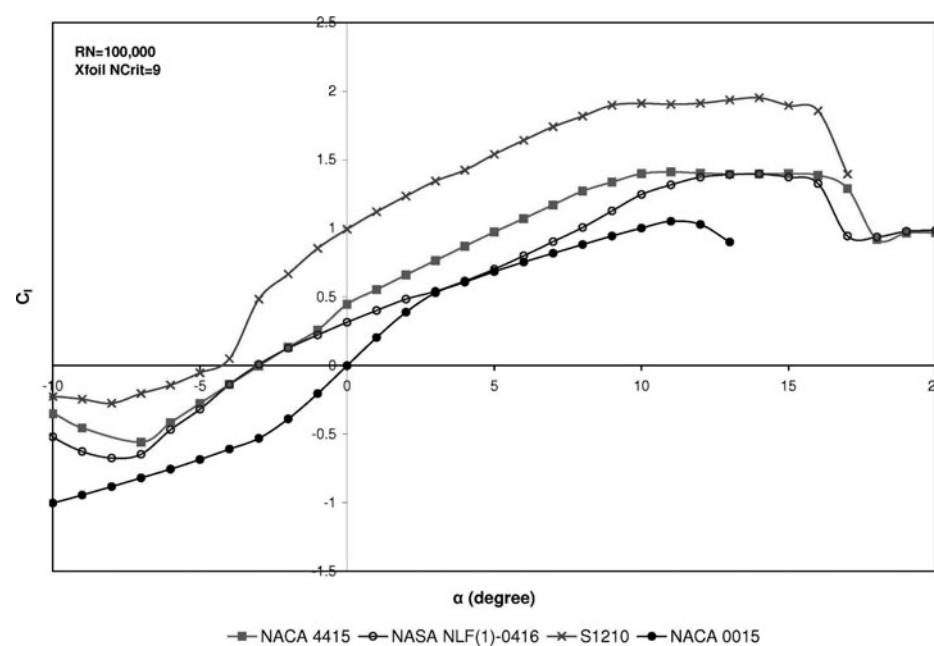


Figure 11. Lift Curves of Selected Asymmetric Airfoils at RN=100,000

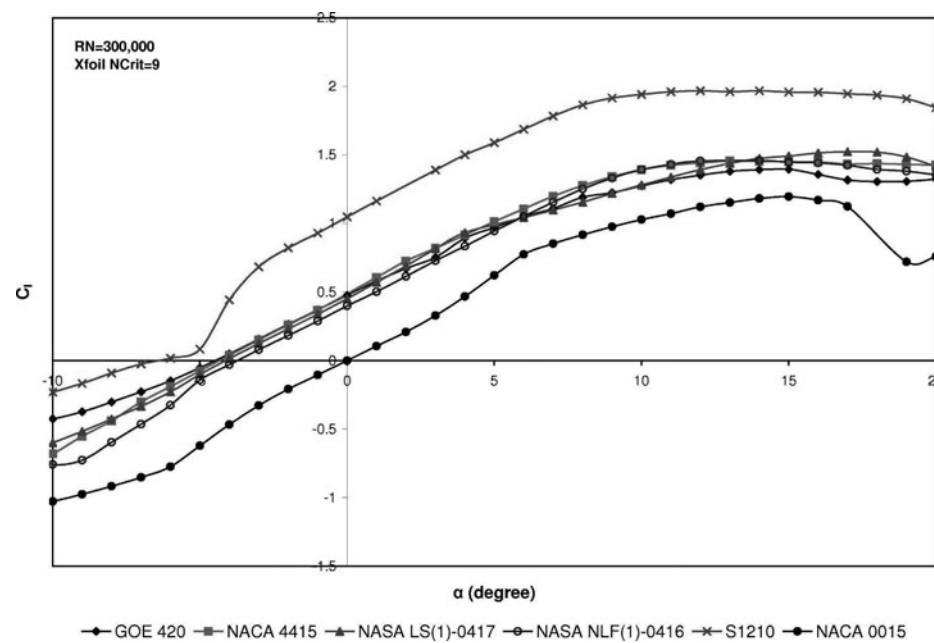


Figure 12. Lift Curves of Selected Asymmetric Airfoils at RN=300,000

6.2. Wide Drag Bucket

Generally, airfoils exhibit the lowest drag over a narrow range of angle of attack called the “drag bucket” and the airfoil shape solely determines the shape and position of the drag bucket [54]. Klimas [2] identified that wide drag bucket is one of the desirable characteristics of VAWTs. It was also commented by Claessens [11] that increased width of drag bucket is required to maintain performance over a larger range of α . In Figures 13 and 14, drag polar ($C_d - C_l$ curves) of selected asymmetric airfoils are shown at RN=100,000 and 300,000. It can be seen in Figure 13, that the width of the drag bucket of NLF-0416 is the largest among all the analyzed airfoils. However, at higher RN of 300,000, as shown in Figure 14, the widths of the drag bucket of LS-0417 and NLF-0416 are similar and not much different from that of NACA 0015.

6.3. Small Zero-lift-drag Coefficient

The desired airfoil should have the least amount of drag, especially the zero-lift-drag coefficient (C_{d0}) should be very low [55, 47]. Seki et. al. [47] also suggested that zero-lift angle and minimum drag coefficient angles of a better performing airfoil should coincide or the difference should be small. Basically, the zero-lift drag coefficient is reflective of parasitic drag which makes it very useful in understanding how “clean” or streamlined a VAWT’s aerodynamics is [56]. It can be observed from Figures 13 and 14 that C_{d0} of NACA 0015 is the lowest among all the airfoils considered. However, it will be shown in section 6.7 that NACA 0015 is very sensitive to roughness.

6.4. Large C_l/C_d ratio

The C_l/C_d ratio indicates the overall efficiency of any airfoil and naturally it is desirable to have higher ratio over a wide range of α for improving the aerodynamic performance of a VAWT [29, 39] by increasing the tangential force coefficient which is defined as:

$$C_T = C_l \sin \alpha - C_d \cos \alpha \quad (1)$$

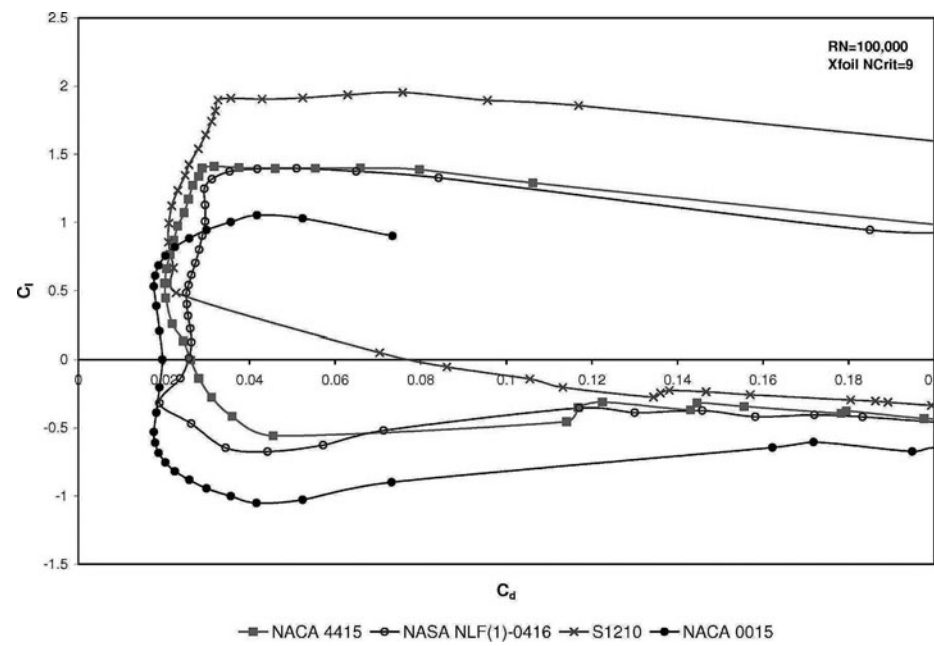


Figure 13. Drag Buckets of Selected Asymmetric Airfoils at RN=100,000

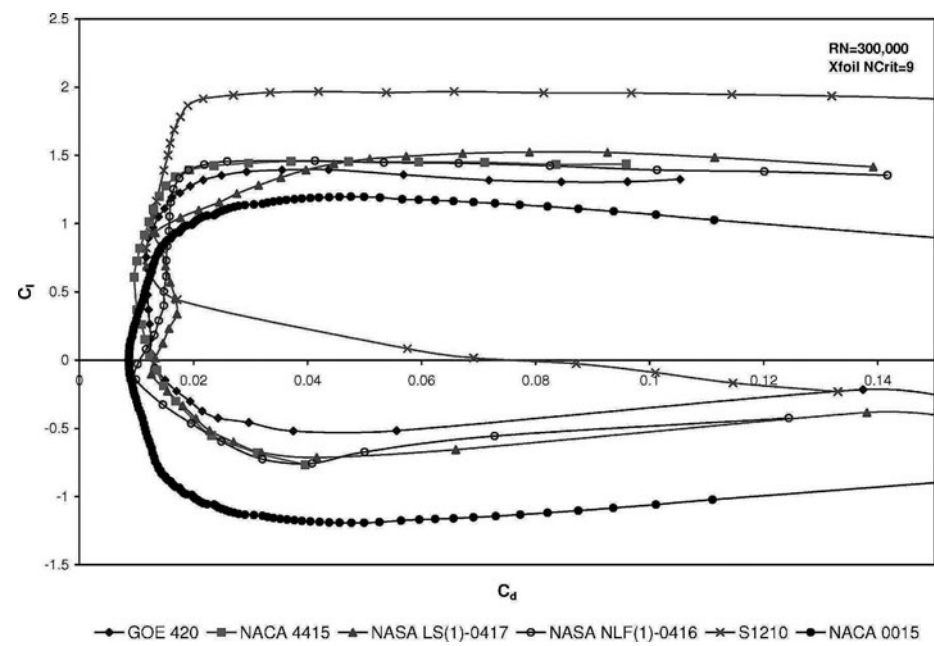
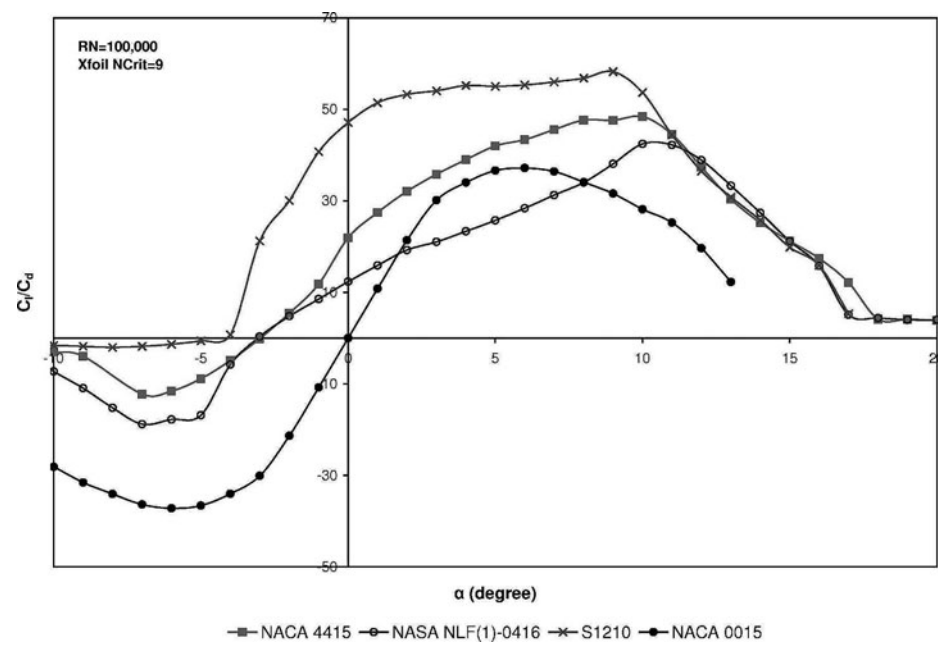
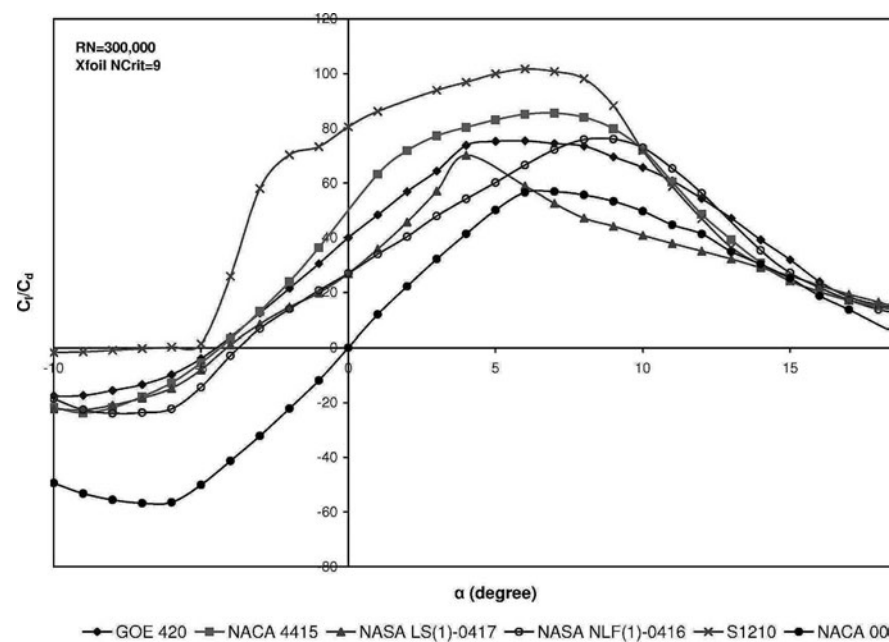


Figure 14. Drag Buckets of Selected Asymmetric Airfoils at RN=300,000

It can be seen in Figure 15 that the C_l/C_d ratio of S1210 and NACA 4415 is much higher over a wide range of positive incidence in the pre-stall range than NACA 0015 at RN=100,000, clearly demonstrating their superiority at positive incidences. However, at negative incidences the reverse is true. But it has already been established in Section 6.1 that relatively the contribution of airfoils at positive incidences is more important than that of negative incidences for the concave-out configuration. At positive incidence and higher RN, the C_l/C_d ratios of all the asymmetric airfoils, except LS-0417, are better than that of NACA 0015, but again the situation is reversed at negative incidences.

Figure 15. C_l/C_d Ratio of Selected Asymmetric Airfoils at $Re=100,000$ Figure 16. C_l/C_d Ratio of Selected Asymmetric Airfoils at $Re=300,000$

6.5. Large Maximum Lift-Coefficient

If the airfoil shape of a smaller-capacity SB-VAWT has higher $C_{l,max}$, more positive torque will be generated in the pre-stall regime. This feature will also enhance the starting torque. Both high lift and stall angle are therefore desirable for VAWTs. However, $C_{l,max}$ should be relatively insensitive to the changes of Re . The $C_{l,max}$ of asymmetric airfoils at positive α are usually much better than the symmetric profiles like NACA 0015 which is depicted in Figures 11 and 12 for Re of 100,000 and 300,000 respectively. Seki et. al. [47] suggested that the slope of the lift curve of SB-VAWT airfoil should be steeper for improved efficiency.

6.6. Delayed Deep-stall Property

As briefly mentioned in Section 2, deep stall has negative influence on the performance of SB-VAWT. According to Claessens [11], deep stall characteristics of airfoil are important for VAWTs and he suggested that: (i) deep stall should be postponed to a larger angle of attack; (ii) hysteresis loop of the deep stall should be as small as possible; and (iii) The drop of lift coefficient should be as small as possible at deep stall. The angle at which deep stall occurs depends on RN and the nose radius. Timmer [57] correlated the available data with the thickness of the airfoil nose, defined as the y/c value at $x/c=0.0125$, following the work of Gault [58], in which the stalling characteristics of a large number of low-speed airfoils have been correlated. Timmer measured the deep stall characteristics of multiple profiles at $RN=1$ million. He found a linear relation between the thickness of the nose and the deep-stall angle and the resulting straight line can be translated to the following relation between the thickness of the nose and the deep stall angle:

$$\alpha_{\text{deep-stall}} = 1114 \left(\frac{y}{c} \right)_{\frac{x}{c}=0.0125} \quad (2)$$

Using the above equation, the authors have investigated the deep-stall characteristics of NACA 0015 and the selected asymmetric airfoils and results are presented in Figures 17 and 18 for the upper and lower surfaces of the airfoils. It is seen in Figure 17 that except NLF-0416, the onset of deep-stall on the upper surface is delayed for the other asymmetric airfoils in comparison to NACA 0015. In Figure 18, the negative values in x/c denote lower surface ordinates and it can be seen that onset of deep-stall of NACA 0015 and GOE 420 are the lowest. It would be worthwhile to highlight that the analysis performed by Timmer [57] is done at high RN range of 1 million and its applicability to low RN range typically encountered by smaller capacity SB-VAWT should be investigated further.

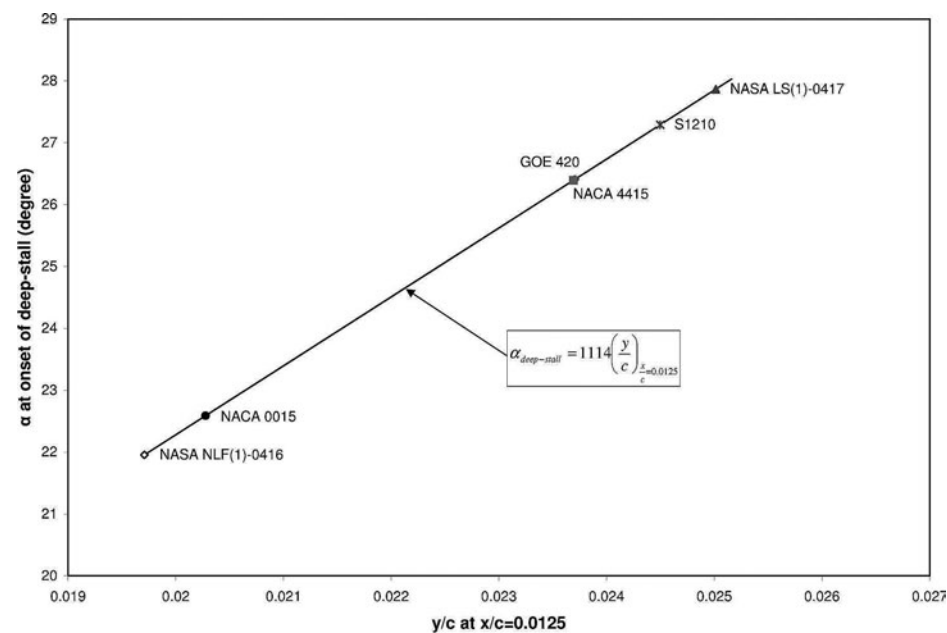


Figure 17. Deep Stall Characteristic on the Upper Surfaces of Selected Prospective Airfoils

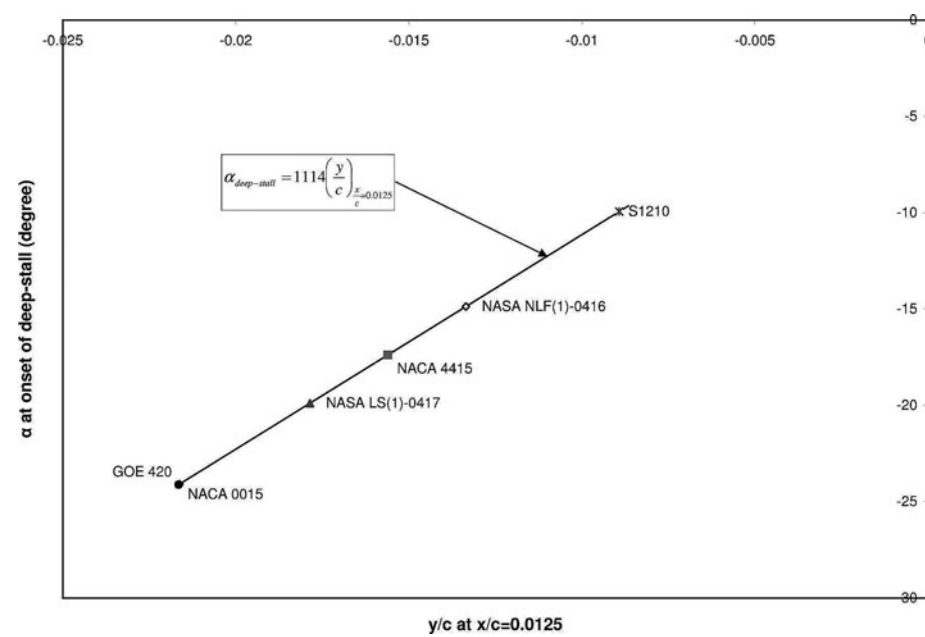


Figure 18. Deep Stall Characteristic on the Lower Surfaces of Selected Prospective Airfoils

6.7. Small Roughness Sensitivity

The airfoil should have least amount of roughness sensitivity as wind turbines operate at diversified climatic conditions and its maintainability and performance deteriorate with surface roughness due to dust, dirt, rain or insect debris [37]. Because of surface roughness the boundary layer of the blades will turn turbulent at the nose, which results in a turbulent boundary layer over the airfoil [11]. Surface roughness generally decreases $C_{l,max}$ and increases C_{do} and these effects become more pronounced as RN increases [25]. According to Lissaman [39], $C_{l,max}$ of the NACA 230XX series airfoils are very sensitive to surface fouling, and their performance deteriorates with increased thickness more rapidly than that of other airfoils. He also commented that NACA 63-2XX series airfoils have demonstrated the best overall performance characteristics of the NACA families, and they give reasonable resistance to roughness losses. Airfoils in the LS(1)-04XX series were designed to tolerate surface fouling, but HAWTs with these airfoils have experienced large power losses induced by roughness [39].

To analyze the roughness sensitivity of airfoils in the wind tunnel, zigzag tape is usually applied at locations near to the leading edge of the airfoil. This situation can be simulated in XFOIL by imposing fixed transition near to the leading edge portion. To compare the roughness sensitivity of the candidate airfoils for SB-VAWT, a trip is used at 10% chord length downstream of the leading edge at $Re=100,000$ and $300,000$. The results of the simulation at 10% fixed transition are compared with the salient characteristics of these airfoils at free transition and they are presented in Tables 1 and 2. It can be seen from Table 1 that for all the airfoils, α_{stall} have increased, whereas $C_{l,max}$ and C_l/C_d ratio have decreased. Mixed trends have been observed for C_{do} and C_m of these airfoils at $Re=100,000$. However, at $Re=300,000$ (as shown in Table 2), α_{stall} and C_{do} of all the airfoils have increased, whereas C_l/C_d and C_m have decreased. In this case, mixed trends have been observed for $C_{l,max}$. It can be inferred from these results that C_{do} , which is a very important parameter for SB-VAWT, is most affected due to roughness and it increases appreciably with RN. Among all the airfoils, C_{do} of LS(1)-0417 is found to be least sensitive to roughness both at $Re=100,000$ and $300,000$ which is consistent with the comment of Lissaman [39] stated above. It is also found that C_{do} of NACA0015 is severely affected at $Re=300,000$ because of roughness.

Table 1. Roughness Sensitivity of Selected Airfoils at RN=100,000													
Airfoil	α_{stall} (degree)		$C_{l_{\text{max}}}$		C_{d0}		C_l/C_d		C_m		Free	Transition	Fixed
	Free	Transition	Free	Transition	Free	Transition	Free	Transition	Free	Transition			
NACA0015	11.00	13	1.05	1.0196	0.0196	0.0204	37.42	28.99	-0.0423	-0.0463			
NACA4415	10.60	14	1.41	1.1994	0.0262	0.0221	48.96	32.01	-0.1073	-0.0798			
LS(1)-0417	12.20	14.6	1.35	1.3052	0.0398	0.0229	42.31	28.87	-0.1238	-0.1263			
NLF-0416	11.20	13.8	1.34	1.1486	0.0258	0.0216	43.13	33.50	-0.1219	-0.1076			
SI210	9.60	14.4	1.91	1.6416	0.0745	0.1171	58.27	45.81	-0.2505	-0.2333			

Table 2. Roughness Sensitivity of Selected Airfoils at RN=300,000													
Airfoil	α_{stall} (degree)		$C_{l_{\text{max}}}$		C_{d0}		C_l/C_d		C_m		Free	Transition	Fixed
	Free	Transition	Free	Transition	Free	Transition	Free	Transition	Free	Transition			
NACA0015	15.00	15.2	1.20	1.2197	0.0086	0.0151	57.75	43.08	-0.0440	-0.0247			
NACA4415	13.00	16.2	1.46	1.3935	0.0126	0.0163	85.59	46.58	-0.1063	-0.0858			
LS(1)-0417	14.80	16.8	1.49	1.5059	0.0130	0.0168	73.76	41.93	-0.1066	-0.0919			
NLF-0416	13.20	16.2	1.47	1.3516	0.0106	0.0159	76.66	50.87	-0.0946	-0.0872			
SI210	11.00	14	1.96	1.843	0.0797	0.1252	101.72	64.25	-0.2456	-0.2319			

6.8. Small Trailing Edge Noise Generation

The laminar separation bubbles that extend over the trailing edge of the airfoil cause the blades to vibrate and are a source of noise. As SB-VAWTs have prospects to operate in the urban environments, noise emission should be kept as low as possible. According to Claessens [11], an airfoil with smooth stall characteristics is desirable to reduce the trailing edge noise. In order to assess the trailing edge noise generated by the candidate airfoils, NAFNoise (which stands for NREL AirFoil Noise) has been utilized in the paper. NAFNoise is a program that predicts the noise of any airfoil shape for different types of noise sources, including trailing-edge (TE) noise [59]. Empirical models for the TE noise sources were originally developed by Brooks, Pope and Marcolini [60] and it models any airfoil shape by using the boundary layer calculations of XFOIL which can be used as inputs into the models of Brooks, Pope, and Marcolini. In Figure 19, sound pressure level (SPL) at 1/3rd octave of candidate airfoils are shown. It can be seen from this figures that among all the airfoils, NLF(1)-0416 and NACA 4415 generate the most and least amounts of noise respectively.

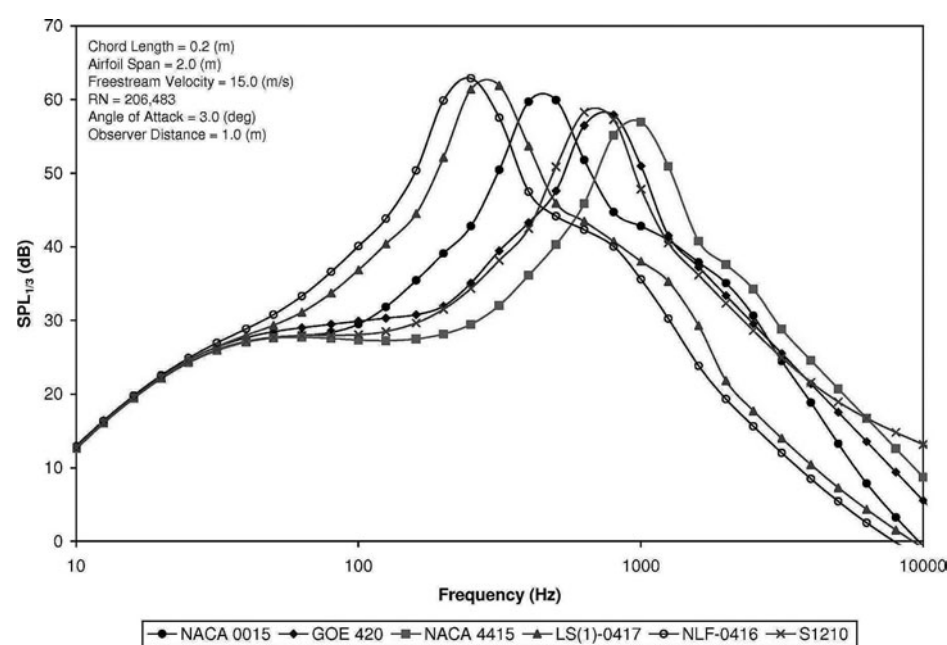


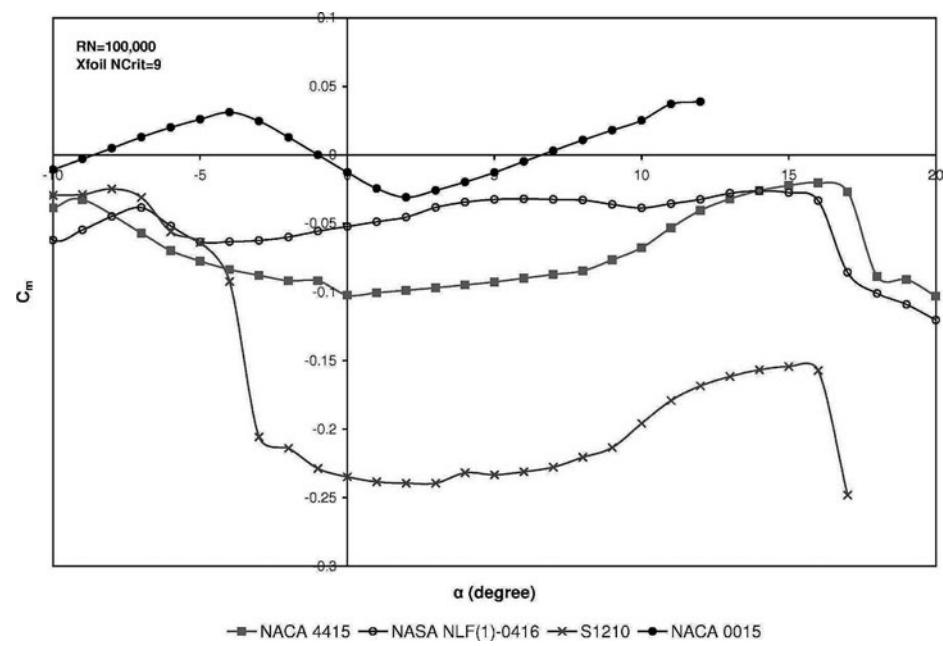
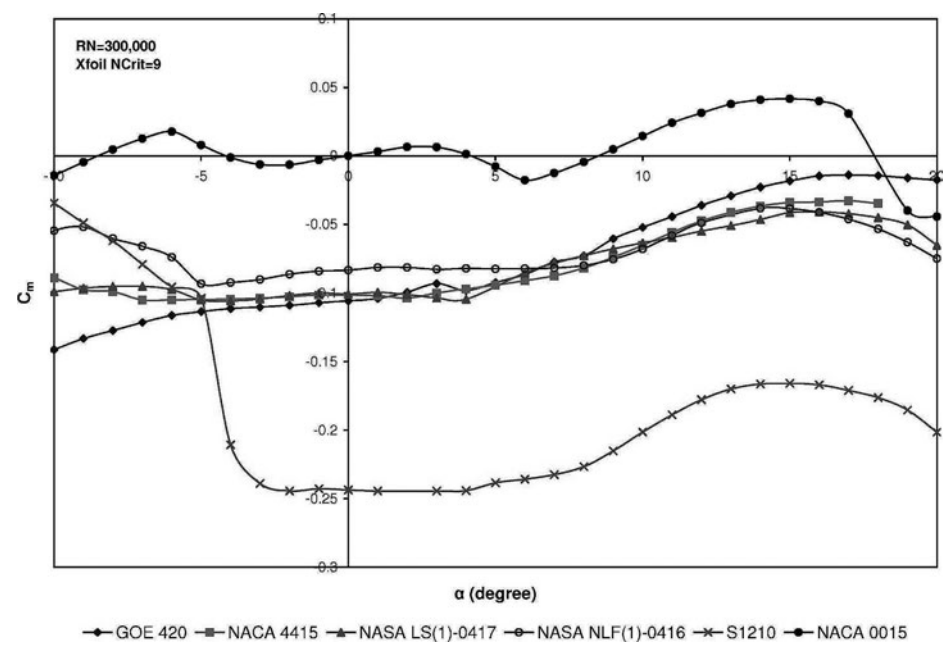
Figure 19. Comparison of Sound Pressure Level (SPL) of Selected Prospective Airfoils

6.9. Large Negative Pitching Moment

Kato et. al. [46] utilized a theory of simple tube of flow and found that pitching moment coefficient (C_m) of a SB-VAWT airfoil should be large for high efficiency. In their publication, they produced C_p - λ curve which shows better performance of an airfoil with $C_m=0.05$ relative to an airfoil with $C_m=0$ at higher TSR (>3.5) range. The variations of C_m of selected airfoils are shown in Figures 20 and 21 for RN of 100,000 and 300,000. For both cases, the values of C_m for S1210 are most negative and they fluctuate around zero for NACA 0015.

In this section, nine desirable aerodynamic characteristics are identified and performance of the selected candidate airfoils is determined. All these characteristics are summarized in Table 3 for overall comparison and each candidate airfoil is rated (as shown within the parenthesis) based on each desirable characteristic for quick reference.

Table 3. Ratings of Candidate Airfoils based on Desirable Aerodynamic Characteristics																
Airfoil	Stall Angle (degree)		Drag Bucket		C_{do}		$(C_l/C_{d_l})_{max}$		$C_{l,max}$		Deep Stall (degree)		Roughness Sensitivity (% Change in C_{do} due to Trip)		$C_{m,min}$	$SP_{L/3,max}$ (dB)
	RN=100,000	RN=300,000	RN=100,000	RN=300,000	RN=100,000	RN=300,000	RN=100,000	RN=300,000	RN=100,000	RN=300,000	Upper Surface	Lower Surface	RN=100,000	RN=300,000	RN=100,000	RN=300,000
NACA 0015	11.0 (3)	15.0 (1)	Large (1)	Large (1)	0.0196 (1)	0.0086 (1)	37.42 (5)	57.75 (6)	1.05 (5)	1.20 (6)	22.59 (5)	-24.12 (1)	4.08 (4)	75.58 (5)	-0.0423 (5)	59.90 (4)
GOE420	-	14.4 (3)	-	Medium (3)	-	0.0127 (4)	-	75.45 (4)	-	1.40 (5)	26.41 (3)	-24.09 (2)	-	-	-0.1496 (2)	57.90 (2)
NACA 4415	10.6 (4)	13.0 (5)	Medium (2)	Medium (2)	0.0258 (2)	0.0126 (3)	48.96 (2)	85.59 (2)	1.41 (2)	1.46 (4)	26.39 (4)	-17.38 (4)	-15.65 (3)	29.37 (2)	-0.1073 (4)	56.95 (1)
LS(1)-0417	12.2 (1)	14.8 (2)	Small (5)	Small (6)	0.0398 (4)	0.013 (5)	42.31 (4)	73.76 (5)	1.35 (3)	1.49 (2)	27.87 (1)	-19.89 (3)	-42.46 (1)	29.23 (1)	-0.1238 (2)	61.92 (5)
NLF(1)- 0416	11.2 (2)	13.2 (4)	Medium (3)	Medium (4)	0.0258 (3)	0.0106 (2)	43.1291 (3)	76.66 (3)	1.34 (4)	1.47 (3)	21.95 (6)	-14.87 (5)	-16.28 (2)	50.00 (3)	-0.1219 (3)	62.88 (6)
S1210	9.6 (5)	11.0 (6)	Medium (4)	Medium (5)	0.0745 (5)	0.0797 (6)	58.2696 (1)	101.72 (1)	1.91 (1)	1.96 (1)	27.29 (2)	-9.94 (6)	57.18 (5)	57.09 (4)	-0.2456 (1)	58.27 (3)

Figure 20. Pitching Moment Coefficient of Selected Asymmetric Airfoils at $Re=100,000$ Figure 21. Pitching Moment Coefficient of Selected Asymmetric Airfoils at $Re=300,000$

7. DESIRABLE GEOMETRIC FEATURES

Desirable aerodynamic characteristics described in the previous section results from the different geometric features of an airfoil as shown in Figure 22. So, it is of profound importance to identify the required geometric features which is done in this section. A designer of small-scale SB-VAWT should consider these features while selecting an airfoil and a judicious choice should be made. Also, performance analysis of a SB-VAWT equipped with the selected airfoil as a blade shape should be done using one of the available computational models [61] before field applications.

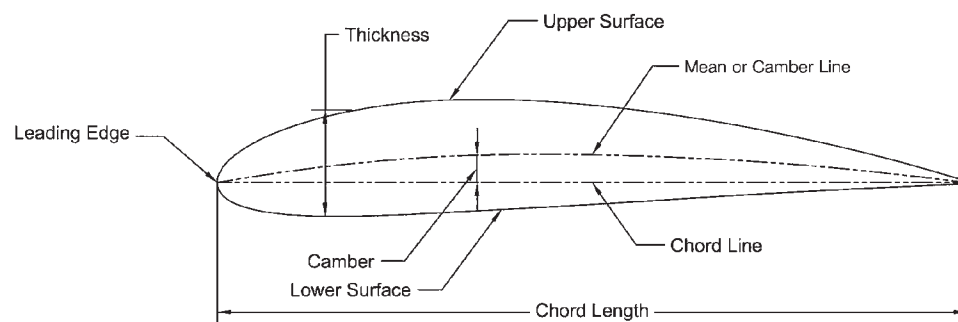


Figure 22. Geometric Features of a Typical Asymmetric Airfoil

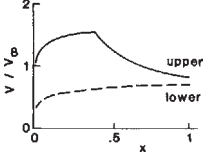
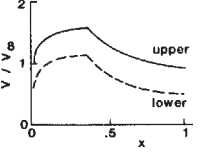
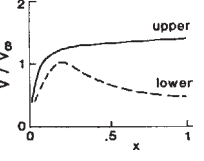
7.1. Camber

Baker [19] compared the theoretical performance of a moderately cambered Göttingen 420 airfoil with that of a symmetrical NACA 0012 airfoil with similar thickness. His modeling indicated that the cambered section would produce a higher tangential thrust (i.e. torque) over a wider range of α and more energy per “cycle” (i.e. per revolution) than the symmetrical section, and that a VAWT using the cambered blades would self-start, unlike the same turbine with the symmetrical blades. Kirke [3] also found that a turbine with cambered NACA 4415 blades of 0.32 m chord should easily self-start in a 10 m/s wind ($Re = 200,000$), unlike an otherwise identical turbine with symmetrical NACA 0015 blades.

Kirke measured and compared the lift curves obtained for three airfoils, namely NACA 0015 (symmetric), NACA 4215 (asymmetric) & NACA 4415 (asymmetric), at three low values of Reynolds number – (i) 85,000; (ii) 124,000; and (iii) 213,000. He found that at the lowest Reynolds number value of 85,000 – “the more highly cambered NACA 4415 section develops much higher lift at positive incidence than the other two sections, while at negative incidence there is little difference between the three curves, i.e. the cambered section is greatly superior at positive incidence and not much inferior at negative incidence and so is likely to give superior starting performance”. He also found that as Re increases, the superiority of the cambered sections in terms of lift at positive incidence becomes less pronounced. The net advantages of asymmetric airfoils over symmetric ones can be summarized as:

- (i) **Larger lift at low Re :** In general, cambered blade sections produce higher lift at low Re and positive incidence than do symmetrical sections. The small-scale SB-VAWTs are usually designed to operate between chord Re of 100,000 to 300,000, which is considered as the Low Re regime. Miley [25] remarked that at low Re , the laminar boundary layer is more stable and resistant to transition. The turbulent boundary layer is weak, and is able to tolerate only mild adverse pressure gradients (APG). There must be a large APG on either the upper or lower surface of an airfoil to generate lift and usually it is done in one of the three ways as described in Table 4. It can be concluded from Miley’s investigation that for low Re operation, the type-3, which is basically a cambered airfoil, produces lift from high pressure on the lower surface and its APG is most moderate of the 3 types. Thus type-3 airfoils are less prone to separation which causes loss of lift and increase in drag.
- (ii) **Less roughness sensitivity:** The type-3 cambered airfoils are also less sensitive to roughness because the behaviour of the boundary layer is less critical for the realization of the lift producing pressure distribution. The increase in drag is common to all the three types of airfoils described above, but for type-3 the loss of lift is generally less [25].

- (iii) **Larger stall angle:** The stall angle of asymmetric cambered airfoils is higher than symmetric airfoils
- (iv) **Large negative pitching moment:** Asymmetric airfoils usually produce large negative pitching moment (unlike symmetric airfoils), they are expected to show better performance in the higher λ range according to the study of Kato et al.[41].

Table 4. Three Types of Airfoils for Generating Lift [25]			
	Type-1	Type-2	Type-3
Geometry			
Velocity distribution			
Adverse pressure gradient (APG)	Requires strong APG on the upper surface	Upper surface requires strong APG (like Type-1), but lower surface required moderate APG.	Require moderate APG.
$C_{l,max}$ as RN decreases	Decreases appreciably	Decreases appreciably	Decreases moderately
α_{Stall} as RN decreases	Decreases appreciably	Decreases appreciably	Decreases moderately
Roughness sensitivity	More sensitive and there is loss of lift and increase in drag	More sensitive and there is loss of lift and increase in drag	Less sensitive and there is loss of lift and increase in drag
Applicability	–	Suitable for aviation applications.	Suitable for Low RN applications and used extensively for flying model aircraft and small low-speed fans.

7.2. Greater Thickness

Blades with thicker airfoils shapes have at least five advantages for small-scale SB-VAWTs which are:

- (i) **Improvement of Performance:** Theoretical work by Healy [62] indicated that thicker sections, at least up to 18%, offer better performance at RN around 200,000-300,000. Also, Angell et. al. [63] showed that sections up to about 20% thickness may be used with no loss in performance at RN of the order of 1.5 million.
- (ii) **Increase in Starting Torque:** They also tend to increase starting torque according to modeling by Simhan [64]. This apparently occurs because the drop in C_l associated with stall occurs less suddenly and sharply than with thinner sections. Dereng [29] suggested that – “Airfoils of 19 percent chordal thickness have shown the most desirable start-up and traverse characteristics.” Seki [44] also suggested a maximum thickness of the 2D blade between 20 and 25 percent of the chord thickness for improving the turbine self-starting performance and the efficiency of his straight wing type wind and water turbine which was patented in the USA.

- (iii) Increase in Width of the Drag Bucket: In some airfoil series, larger thickness makes a wider drag bucket to maintain performance over a larger range of α [11].
- (iv) Lower Radiated Sound: According to experiments conducted at Dutch national aerospace laboratory, It has been found out that the radiated sound decreases with an increasing blade thickness [65].
- (v) Improvement of Structural Strength: Blades with thicker airfoil shape can be made structurally stronger [37] with little increase in cost.

These above-mentioned advantages are due to several factors, including (i) soft stall or trailing edge stall which delays separation; (ii) higher amount of torque and (iii) wide range for $C_{l,max}$ [3, 62]. It has also been identified by Claessens [11] that a thicker airfoil will in general also have deep stall at a higher angle of attack and they have generally a larger drop in lift at deep stall. It was also pointed out by Zervos [22] that the effect of thickness on detrimental unsteady loading was found to be very small. However, there is a limit to the benefits to be gained by using very thick airfoils at very low RN. Sato and Sunada [66] has shown that airfoils of 28.5% thickness may fail to develop any useful lift at very low RN of the order of 33,000 and although RN right down to zero are encountered by VAWT blades moving downwind at λ near 1, the aerodynamic force on these blades moving downwind at low relative velocity (W) is small compared with that on blades moving upwind so the effect of poor performance of blades at very low RN is small [3]. To elucidate the performance of a low-speed asymmetric airfoil at different thickness, C_t - α curves of NASA LS-0417MOD and LS-0421MOD at RN=100,000 and 300,000 are shown in Figures 23 and 24 respectively. We can see from Figure 23 that C_t values of NASA LS-0421 is higher in the post-stall regimes at RN=100,000. At RN=300,000, as shown in Figure 24, the C_t values of both the airfoils are similar.

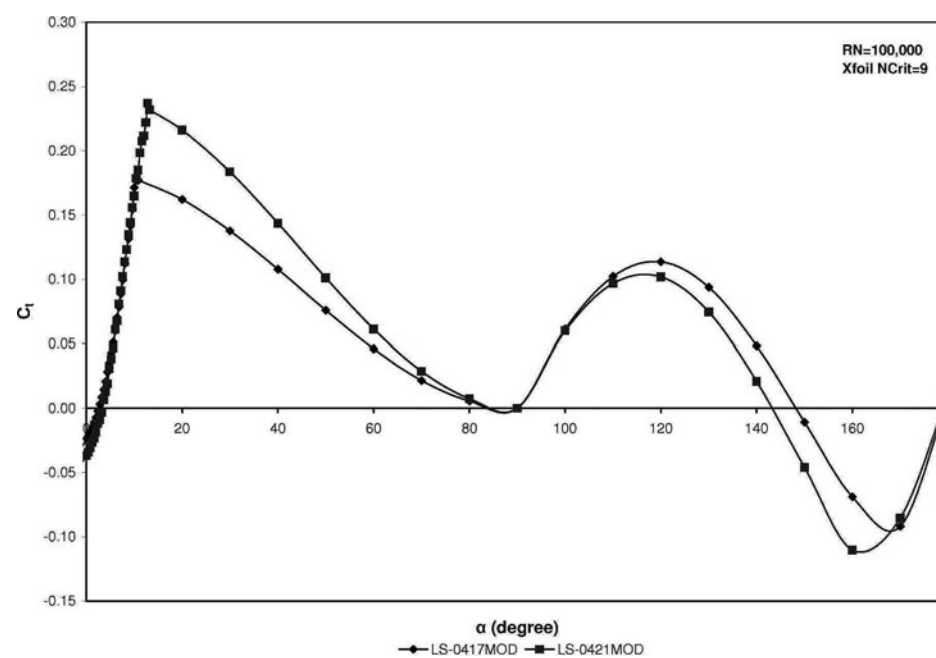


Figure 23. C_t - α curves of NASA LS-0417MOD and LS-0421MOD at RN=100,000

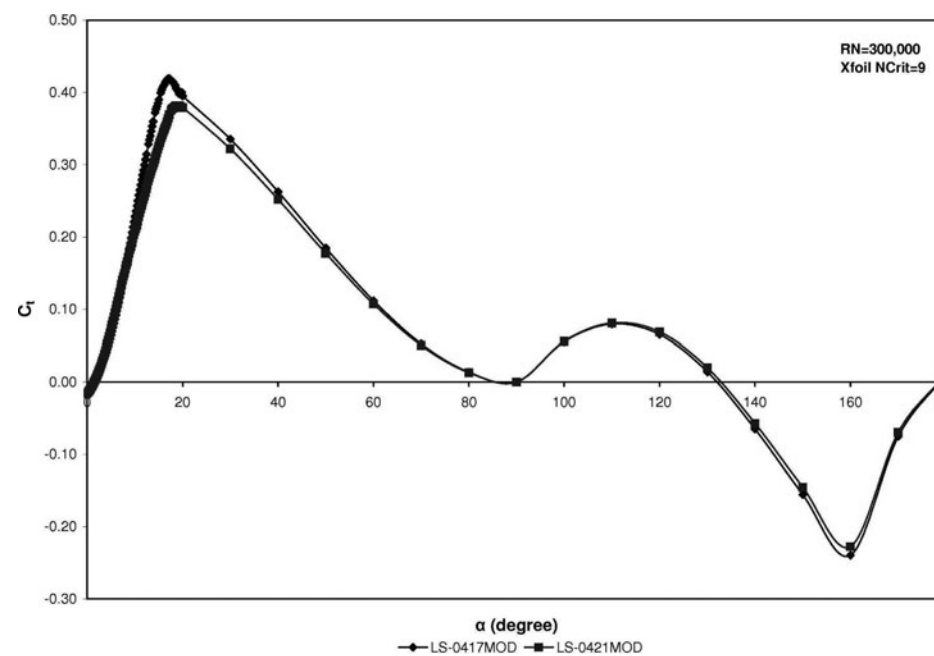


Figure 24. C_t - α curves of NASA LS-0417MOD and LS-0421MOD at $Re=300,000$

7.3. Large Leading Edge Radius

Fupeng et. al. [67] suggested that larger leading edge radius is desirable as they are less sensitive to roughness which is a desirable aerodynamic characteristic for smaller capacity SB-VAWT as mentioned in Section 6.7. So, larger leading edge radius will result in improved energy harness by SB-VAWT under dirty blade conditions owing to the accumulation of insect debris and also to increases of the lift-drag ratio of airfoil at large angle of attack. As described in Section 6.6, deep stall characteristic of an airfoil also depends on leading edge radius. Dereng [29] suggested well rounded airfoils for VAWTs.

7.4. Sharp Trailing Edge

A study of trailing edge treatment was carried out by Sandia National Laboratory on SAND 0018/50, an airfoil specifically designed for VAWT, using extruded aluminum alloy model and it was found out that sharp trailing edge significantly reduced the minimum drag coefficient [68]. Dereng [29] also suggested that – “a well rounded leading edge in combination with a trailing edge which is thin compared to the leading edge, such leading and trailing edges being connected by smooth continuous contours of thickness distribution most conducive to high lift to drag ratio”. However, it should be pointed out that fabrication of sharp trailing edges is often quite difficult and costly.

For comparison purpose, the salient geometric features of the selected candidate airfoils that have been analyzed in the previous section are presented and rated according to the findings of this section in Table 5. However, it should be mentioned that the optimum camber for self-starting and better performing SB-VAWT is related to chord-radius ratios because of flow curvature effects and it should be judiciously chosen.

Table 5. Geometric Features of Selected Airfoils				
Airfoil	Camber (% of c)	Thickness (% of c)	Leading Edge Radius (% of c)	Trailing Edge Thickness (% of c)
NACA 0015	0.00 (6)	15.00 (4)	2.37 (3)	0.32 (2)
GOE420	4.31 (2)	18.76 (1)	2.70 (2)	0.00 (1)
NACA 4415	4.00 (3)	15.00 (4)	2.22 (4)	0.32 (2)
LS(1)-0417	2.17 (5)	16.98 (2)	2.97 (1)	0.71 (3)
NLF(1)-0416	2.44 (4)	15.95 (3)	1.36 (6)	0.00 (1)
S1210	7.20 (1)	11.99 (5)	1.80 (5)	0.00 (1)

8. CONCLUSIONS

The aim of this paper was to identify the desirable features of an ideal airfoil for smaller capacity SB-VAWT to improve its starting characteristics and overall performance. An attempt was made to shortlist several aerodynamic characteristics of the desirable airfoil and subsequently the required geometric features to achieve the short listed characteristics were also determined. It has been found out in this paper that conventionally used old NACA 4-digit symmetric airfoils are not suitable for smaller capacity SB-VAWTs. Rather, it will be advantageous to utilize a high-lift and low-drag asymmetric thick airfoil suitable for low speed operation typically encountered by SB-VAWT.

Selection of the Airfoil is crucial for better aerodynamic performance and dimensions of a smaller-capacity SB-VAWT which can compete with conventional energy sources in niche markets like urban areas and off-grid remote applications for diversified applications. However, in order to install SB-VAWT for numerous prospective applications, a robust design with self-starting characteristics and optimum performance is a must. In this paper, nine aerodynamic features which would contribute to self-starting are identified, these are: (i) high stall angle at low Reynolds number, (ii) wide drag bucket, (iii) low zero-lift-drag coefficient, (iv) high C_l/C_d ratio, (v) high maximum lift-coefficient, (vi) delayed deep-stall property, (vii) low roughness sensitivity, (viii) low trailing edge noise generation, and (ix) large negative pitching moment. After identification of these aerodynamic features, attempt was made to select the desirable geometric features. It has been found that an airfoil for self-starting SB-VAWT with optimum performance should have camber, high thickness, large leading edge radius and sharp trailing edge.

Based on literature survey five asymmetric airfoils, designed mainly for the aviation purposes, found in the public domain were identified as potential candidate for SB-VAWTs by other researchers in the past. These candidate airfoils are investigated in this paper and it has been found that none of these airfoils has all the desirable characteristics or geometric features. Four special-purpose asymmetric airfoils, designed for SB-VAWTs, have also been identified from the literature but could not be investigated due to non-availability of their geometry and copyright related issues. However, none of these special-purpose airfoils seems to be designed for the low RN operation and their main emphasis was on better performance. Under these backdrops, there is a clear need for designing special-purpose airfoils in the light of desirable characteristics and geometric features identified in this paper for self-starting for SB-VAWTs with optimum performance at low RN.

9. ACKNOWLEDGEMENTS

The authors would also like to acknowledge the works of the individuals and organizations that are listed in the following reference section.

10. REFERENCES

1. Iida, A., Mizuno, A., *Prediction Of Aerodynamic Noise Radiated From A Vertical-Axis Wind Turbine*, Proceedings of the 4th ASME/JSME Joint Fluids Engineering Conference, Hawaii, USA, July 6-10, 2003.
2. Klimas, P.C., *Airfoil treatments for vertical axis wind turbines*. WINDPOWER '85, San Francisco, CA, USA. 27 August, 1985.
3. Kirke, B.K., *Evaluation of self-starting vertical axis wind turbines for stand-alone applications*, PhD Thesis, Griffith University, Australia, 1998.
4. Sahin, A. D., *Progress and recent trends in wind energy*, Progress in Energy and Combustion Science, 2004, 30(5), 501-543.
5. Hak, M.G-E., *Control of Low-Reynolds-Number Airfoils – A Review*. Lecture Notes in Engineering, 1989, 246-270.
6. Tani, I., *Low Speed Flows Involving Bubble Separations*, Progress in Aeronautical Sciences, 1964, 5(70), 70-103.
7. Hak, M.G-E., *Control of Low-Speed Airfoil Aerodynamics*, AIAA Journal, 1990, 28, 1537-1552.
8. Migliore, P.G., Wolfe, W.P. and Fanucci, J. B. *Flow Curvature Effects on Darrieus Turbine Blade Aerodynamics*, Journal of Energy, 1980, 4(2), 49-55.
9. Mandal, A.C. and J.D. Burton., *The Effects of Dynamic Stall and Flow Curvature on the Aerodynamics of Darrieus Turbines Applying the Cascade Model*, Wind Engineering, 1994, 18(6), 267-282.
10. Hirsch, C. and Mandal, A.C., *Flow Curvature Effect on Vertical Axis Darrieus Wind Turbine Having High Chord-Radius Ratio*, Proceedings of European Wind Energy Conference, Hamburg, 22-26. October, 1984, 405-410.
11. Claessens, M.C., *The Design and Testing of Airfoils for Application in Small Vertical Axis Wind Turbines*, Master of Science Thesis, Faculty of Aerospace Engineering, Delft University of Technology, The Netherlands, November. 2006.
12. Hoerner, S.F. and Borst, H.V., *Fluid Dynamic Lift, Hoerner Fluid Dynamics*, P.O. Box 21992, Bakersfield, CA 93390, 1985.
13. Klimas, P.C., *Darrieus rotor aerodynamics*. ASME Transactions, Journal of Solar Energy Engineering, 1982, 104, 102-105.
14. Pawsey, N.C.K., *Development and Evaluation of Passive Variable-pitch Vertical Axis Wind Turbines*, Doctoral Thesis, The University of New South Wales. November, 2002.
15. Watson, G. R., *The Self Starting Capabilities of Low Solidity Fixed Pitch Darrieus Rotor*, 1st British Wind Energy Association Workshop paper, 1979, 32-39.
16. Hirsch, C. and Mandal, A.C. *The cascade theory for the aerodynamic performance of Darrieus turbines*, Wind Engineering, 1987, 11, 164-175.
17. Sheldahl, R. E. and Klimas, P. C. (1981). *Aerodynamic Characteristics of Seven Symmetrical Airfoil Sections Through 180-degree Angle of Attack for Use in Aerodynamic Analysis of Vertical Axis Wind Turbines*, Sandia National Laboratories, Report SAND80-2114.

18. Cyberiad, 2007, *Airfoil Data*, web: <http://www.cyberiad.net/foildata.htm>. Last visited on May 17, 2007.
19. Baker, J.R., *Features to aid or enable self-starting of fixed pitch low solidity vertical axis wind turbines*, Journal of Wind Energy & Industrial Aerodynamics, 1983, 15, 369-380.
20. Kentfield, J.A.C., *The Fundamentals of Wind-driven Water Pumps*, Gordon and Breach Science Publishers, 1996.
21. Islam, M., Ting, D. S-K. and Fartaj, A. *Assessment of Small-Capacity Straight-bladed VAWT for Sustainable Development of Canada*. International Journal of Environment Studies: Special Canadian Issue on Energy Conversion and the Environment. 2007, 64(4), 489-500.
22. Zervos, A., *Aerodynamic Evaluation of Blade Profile for Vertical Axis Wind Turbines*. European Wind Energy Conference, Herning, Denmark, 1988, June 6-10. 611-616.
23. Migliore, P.G., *Comparison of NACA 6-Series and 4-Digit Airfoils for Darrieus Wind Turbine*, Journal of Energy, 1983, 7(4), 291-292.
24. Healy, J.V., *The Influence of Blade Camber on the Output of Vertical Axis Wind Turbines*, Wind Engineering, 1978, 2(3), 146-155.
25. Miley, S.J., *A Catalog of Low Reynolds Number Airfoil Data for Wind Turbine Applications*, Department of Aerospace Engineering, Texas A&M University, 1982.
26. Kaldlec, E. G. *Characteristics of Future Vertical Axis Wind Turbines*, Sandia National Laboratory, New Mexico, USA. November, 1982.
27. Zervos, A., *Aerodynamic Design and Testing of Blade Profiles for Vertical Axis Wind Turbines*, Commission of the European Communities. Contractors' meeting, Brussels, 1988. 309-320
28. Zervos, A., *Aerodynamic Evaluation of Blade Profile for Vertical Axis Wind Turbines*, European Wind Energy Conference, Herning, Denmark. June 6-10, 1988, 611-616.
29. Dereng, V.G., *Fixed Geometry Self Starting Transverse Axis Wind Turbine*, United States Patent, Patent No. 4,264,279, April 28, 1981.
30. XFOIL: *Subsonic Airfoil Development System*, web: <http://web.mit.edu/drela/Public/web/xfoil/>. Last visited on May 31, 2007
31. Drela, M. XFOIL: *An Analysis and Design System for Low Reynolds Number Airfoils* (XFOIL 6.94, 18 December 2001), Low Reynolds Number Aerodynamics, T. J. Mueller (ed.), Lecture Notes in Engineering, Springer-Verlag, Berlin, 1989, 54, 1-12.
32. Somers, D.M. and Maughmer, M.D., *Theoretical Aerodynamic Analyses of Six Airfoils for Use on Small Wind Turbines*, Subcontractor Report. National Renewable Energy Laboratory, USA, 2002
33. Dahl, K.S. and Fuglsang, P., *Design of the Wind Turbine Airfoil Family RISØ-A-XX*, Risø National Laboratory, Roskilde, Denmark. December, 1998.
34. Dahl, K.S., *Experimental Verifications of the New RISØ-A1 Airfoil Family for Wind Turbines*, Proceedings of European Wind Energy Conference, 1999. 85-88.
35. Timmer, W.A. and van Rooy, R.P.J.O.M., *Thick airfoils for HAWTs*, Journal of Wind Engineering and Industrial Aerodynamics, 1992, 39(1-3), 151-60.
36. Viterna, L.A. and Corrigan, R.D., *Fixed pitch rotor performance of large wind turbines*, DOE/NASA Workshop on Large Horizontal Axis Wind Turbines, July, 1981.

37. Hansen, A. C. and Butterfield, C. P., *Aerodynamics of Horizontal-Axis Wind Turbines*, Annual Review of Fluid Mechanics, 1993, 25, 115-149.
38. Jonkman, J.M., *Modeling of the UAE Wind Turbine for Refinement of FAST_AD*, NREL Technical Report. NREL/TP-500-34755. December, 2003.
39. Lissaman, P.B.S., *Wind Turbine Airfoils and Rotor Wakes*, Wind Turbine Technology, ASME Press, New York, USA, 1994, 283-323.
40. NWTC *Design Codes*, FoilCheck, 2007. Web: <http://wind.nrel.gov/designcodes/preprocessors/foilchk/>. Last visited on June 8, 2007.
41. AeroDyn, *User's Guide to the Wind Turbine Aerodynamics Computer Software*, Windward Engineering, 2002.
42. Devinant, Ph., Laverne, T., Hureau, J. (2002). *Experimental study of wind-turbine airfoil aerodynamics in high turbulence*, Journal of Wind Engineering & Industrial Aerodynamics, 90, 689 - 707.
43. Kato, Y., Seki, K., and Shimizu, Y., *Vertical Axis Type Wind Power Turbine*, United States Patent, Patent No. 4,285,636, August 25, 1981.
44. Seki, K., *Straight Wind Type Wind and Water Turbine*, United States Patent, Patent No. 6,974,309 B2, December 13, 2005.
45. Seki, K., Shimizu, Y., and Matsumoto, T., *Study of Straight Blade VAWT And Its Application*, European Wind Energy Conference, 1985, 436-442.
46. Kato, Y., Seki, K., and Shimizu, Y., *Vertical Axis Wind Turbine Designed Aerodynamically At Tokai University*, Periodica Polytechnica, Mechanical Engineering, 1981, 25(1), 47-56.
47. Seki, K., Shimizu, Y. and Oshishi, T., *A Study of Aerodynamic Performance and Airfoil for Straight Blade Non-Articulated Vertical Axis Wind and Its Application*, Wind Power 85, 1985, 531-536.
48. Zervos, A., *Aerodynamic Design of Blade Profiles for Vertical Axis Wind Turbines*, Euroforum, New Energies, Proceedings of an international congress held at Saarbrücken, FR. Germany, October, 1988, 24-28.
49. Zervos, A., Morfiadakis, V. and Stefanatos, N., *Testing of blade profiles for vertical axis wind turbines*, Proceedings of European Wind Energy Conference (EWEC) 1989, 193-197.
50. Zervos, A. and Morfiadakis, E., *Instantaneous Pressure Distribution Measurements on The Blades of A Vertical Axis Wind Turbine*, European Wind Energy Conference, 1990, 247-251.
51. Maruyama, Y. Shimura, M., Yoshie, R., Wei, R., and Seki, K., *Development of Vertical Axis Wind Turbine with Straight Blades Suitable for Buildings*, European Wind Energy Conference, 2001, 530-533.
52. Duremberg, C. J., *Unsteady aerodynamics of Vertical Axis Wind Turbines*. 1st British Wind Energy Association workshop, 1979.
53. Loth, J.L. and McCoy, H., *Optimization of Darrieus turbines with an upwind and downwind momentum model*. Journal of Energy, 1983, 7(4), 313-318.
54. The Charles River Radio Controllers (CRRC), *Airfoil Performance Concepts*, 2007. web: http://www.charlesriverrc.org/articles/asfwpp/lelke_airfoilperf.htm. Last visited on May 30, 2007,

55. Jesch, L.F. and Walton, D., *Reynolds Number Effects on the Aerodynamic Performance of a Vertical Axis Wind Turbine*, Proceedings of 3rd International Symposium on Wind Energy Systems, Lyngby, Denmark, August 26-29, 1980, 323-332.
56. Wikipedia. *Zero-lift drag coefficient*, web: http://en.wikipedia.org/wiki/Zero-lift_drag_coefficient, accessed on March 12, 2007.
57. Timmer, W.A., *Summary of The Delft University Wind Turbine Dedicated Airfoils*, 41st AIAA Aerospace Sciences Meeting and Exhibit, 2003, 11-21.
58. Gault, D.E., *A Correlation of Low-Speed Airfoil Section Stalling Characteristics with Reynolds Number And Airfoil Geometry*, NACA Technical Note 3963, Washington. March, 1957.
59. Moriarty, P. NREL, *NAFNoise User's Guide*, NREL-NWTC, Colorado, USA, July 2005.
60. Brooks, T., Pope, D., and Marcolini, M., *Airfoil Self-Noise and Prediction*, NASA Reference Publication 1218, National Aeronautics and Space Administration, 1989.
61. Islam, M., Ting, D. S-K. and Fartaj, A., *Aerodynamic Models for Darrieus Type Straight-bladed Vertical Axis Wind Turbines*, Renewable & Sustainable Energy Review, 2007, In Press.
62. Healy, J.V., *The Influence of Blade Thickness on the Output of Vertical Axis Wind Turbines*, Wind Engineering, 1978, 2(1), 1-9.
63. Angell, R.K., Musgrove, P.J. and Galbraith, R.A.McD., *Unsteady wind tunnel testing of thick section aerofoils for use on large scale vertical axis wind turbines*, Proceeding of 10th BWEA Conference, London, UK, March 22-24, 1988, 195-203.
64. Simhan, K., *A review of calculation methods for the determination of performance characteristics of vertical axis wind energy converters with special reference to the influence of solidity on starting characteristics*. Proceedings of European Wind Energy Conference, Hamburg, FRG, October, 1994, 324-331.
65. Parchen, R., Bruggeman, J.C. and Dassen, A.G.M., *The Effect Of The Blade Thickness Of Wind Turbine Blades On The Noise Due To Inflow Turbulence*, Noise And Vibration Bulletin, July, 1997.
66. Sato, J., and Sunada, Y., *Experimental Research on Blunt Trailing Edge Airfoil Sections at Low Reynolds Numbers*, AIAA Journal, 1995, 33(11), 2001-2005.
67. Fupeng, H., Yuhong, L., and Zuoyi, C., *Suggestions for Improving Wind Turbines Blade Characteristics*, Wind Engineering, 2001, 25(2), 105-113.
68. Paraschivoiu, I., *Wind Turbine Design: With Emphasis on Darrieus Concept*, Polytechnic International Press, Montreal, Canada, 2002.